

Design of an Operational Amplifier for High Performance Pipelined ADCs in 65nm CMOS

Master thesis performed in Electronic Devices

Author: Sima Payami

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Linköping, June 2012

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Master thesis Performed in Electronic Devices
at Linköping Institute of Technology
by Sima Payami

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Keywords Pipelined, ADC, OpAmp, Gain Boosting, CMFB, 2.5bps architecture, Flash, MDAC

Abstract

In this work, a fully differential Operational Amplifier (OpAmp) with high Gain-Bandwidth (GBW), high linearity and Signal-to-Noise ratio (SNR) has been designed in 65nm CMOS technology with 1.1v supply voltage. The performance of the OpAmp is evaluated using Cadence and Matlab simulations and it satisfies the stringent requirements on the amplifier to be used in a 12-bit pipelined ADC. The open-loop DC-gain of the OpAmp is 72.35 dB with unity-frequency of 4.077 GHz. Phase-Margin (PM) of the amplifier is equal to 76 degree. Applying maximum input swing to the amplifier, it settles within 0.5 LSB error of its final value in less than 4.5 ns. SNR value of the OpAmp is calculated for different input frequencies and amplitudes and it stays above 100 dB for frequencies up to 320MHz.

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Keywords: Pipelined, ADC, OpAmp, Gain Boosting, CMFB, 2.5bps architecture, Flash, MDAC

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Introduction

Overview

Analogue to digital converters are the most important building blocks in lots of applications. As electronics and telecommunication worlds are moving fast towards digitalization and there is an ever increasing demand on speed and accuracy of the processed data, the need for high speed and high resolution ADCs has grown dramatically over recent years. There are many types of ADCs that one can choose between them, but based on the application specification and the requirements on speed, resolution, power and area the most suitable architecture can be chosen.

For high speed and medium resolution (10-12 bits), pipelined ADCs are the architecture of choice in most cases. Pipelined ADC falls in the category of multi-stage ADCs which hire stages with lower resolution and resolve more bits by using several stages rather than by incorporating one high resolution ADC. In this way the speed and accuracy requirements on separate stages decrease. Each stage of the pipelined ADC includes a low resolution flash ADC and a Multiplying DAC (MDAC). The flash ADC resolves a few bits from an input sample and the MDAC is responsible for reconstructing these bits into analogue sample, comparing it to the input sample, generating an error signal and amplifying the error signal to be applied to the next stage. The amplification in the MDAC is done using an Operational Amplifier (OpAmp) placed in a feedback system which provides closed-loop feed-forward gain of 2^m , in which m is the stage resolution.

OpAmps are basic building blocks of a wide range of analogue and mixed signal systems. Basically, OpAmps are voltage amplifiers being used for achieving high gain by applying differential inputs. The gain is typically between 50 to 60 decibels. This means that even very small voltage difference between the input terminals drives the output voltage to the supply voltage. In the case of using 65nm CMOS technology, this small voltage difference can be around tens of millivolts. As new generations of CMOS technology tend to have shorter transistor channel length and scaled down supply voltage, the design of OpAmps stays a challenge for designers.

For a 12-bit pipelined ADC with sampling rates higher than 50MS/s, the requirements on the OpAmp are high. The OpAmp should be designed such that to provide high Gain – Bandwidth (GBW), fast settling, high linearity and good enough noise response to satisfy those requirements. For example a GBW of around 2GHz is required for 12-bit pipelined ADC with 3-bit resolution in each stage and sampling frequency of 300MHz. These high requirements are getting harder to achieve as new technologies are scaling down continuously. Recently published works about ADCs employ more complex digital correction circuitry and calibration techniques and focus on finding new solutions to avoid the problems accompanying OpAmp-based designs. Nevertheless, design of the OpAmps, with the aim of making improvements to their performance metrics, is still a worthy field of research.

In this work, an OpAmp with high gain-bandwidth, high linearity and SNR has been designed. The performance of the OpAmp is calculated using Cadence and Matlab simulations and they satisfy the requirements on the high performance amplifier needed in a 12-bit pipelined ADC. The open-loop DC-gain of the OpAmp is 72.35 dB with unity-frequency of 4.077 GHz. Phase-Margin (PM) of the amplifier is equal to 76 degree. Applying maximum input swing to the amplifier, it settles within 0.5 LSB error of its final value in less than 4.5 ns. SNR value of the OpAmp is calculated for different input

frequencies and amplitudes and its value stays above 100 dB for frequencies up to 320MHz.

The amplifier is placed in a pipelined ADC which is also designed in transistor level to check for its functionality. The main focus in this work is the OpAmp design to meet the stringent requirements needed for the 12-bit pipelined ADC. The OpAmp provides enough closed-loop bandwidth to accommodate a high speed ADC (around 300MSPS) with very low gain error to match the accuracy of the 12-bit resolution ADC.

Thesis Organisation

In Chapter1, different ADC architectures (SAR, folding, flash, sub-ranging and $\Sigma\Delta$ ADCs) are briefly discussed. Afterwards, the ADCs' error sources, the definition of their static and dynamic errors and the standard performance metrics to quantify these errors are described.

In Chapter2, the pipelined ADC's architecture is shown. Then the transistor level circuits of its building blocks such as comparator, resistive ladder DAC, thermometer decoder, switched capacitor sampling network, bootstrap circuit for sampling switches, etc. are displayed and their design considerations are discussed.

In Chapter3, ideal and non-ideal OpAmps and their properties are shown and discussed. Then, OpAmp's different topologies are presented. These topologies are telescopic topology, folded-cascode topology, two-stage OpAmps and gain boosted OpAmps. At the end these topologies are compared against each other.

In Chapter4, necessary requirements for an OpAmp to be used in a 12-bit pipelined ADC, with 2.5 bit-per-stage (bps) stage architecture, are calculated. Then the designed OpAmp is presented and the OpAmp's simulated performance is depicted.

In Chapter5, simulation results of the pipelined ADC are shown. Two models of pipelined ADC are introduced and their simulation results are illustrated. First model is a completely high level pipelined ADC with all blocks in VerilogA code. The high level model's simulation result is a very convenient reference to be compared with the other model's performance. The second model is similar to the high level pipelined ADC except for the inter-stage gain provider which is replaced with the designed OpAmp in a closed-loop configuration with feed-forward gain of 4. This model is used to verify the OpAmp's performance in the ADC's circuit.

In Future Work section, some areas that are not covered in this thesis are recommended to continue this work. Research areas that are proposed include power optimization, digital calibration and time interleaving.

In Appendix A, the simulation result for the completely transistor level pipelined ADC introduced in Chapter2 is illustrated. The performance metrics of this model are calculated for different sampling frequencies and peak-to-peak differential voltage amplitudes of input signal like the other two models in Chapter5.

In Appendix B, VerilogA and Matlab codes which are used in this thesis are presented. VerilogA codes are responsible for sampling the output signal of the OpAmp and digital output bits of the ADC and dump them into a text file which can be used by Matlab codes to reconstruct the digital bits and calculate the performance metrics.

List of Acronyms

Bellow, acronyms used in this thesis are listed:

$\Sigma\text{-}\Delta$	Sigma-Delta Analogue to Digital Converter
ADC	Analogue to Digital Converter
bps	bit per stage
CM	Common Mode
CMFB	Common Mode Feed Back
CMRR	Common Mode Rejection Ratio
DAC	Digital to Analogue Converter
DNL	Differential Non Linearity
ENOB	Effective Number of Bits
GBW	Gain Bandwidth
INL	Integral Non Linearity
LSB	Least Significant Bit
MDAC	Multiplying DAC
MSB	Most Significant Bit
OpAmp	Operational Amplifier
PM	Phase-Margin
rms	root mean square
SAR	Successive Approximation Register
SFDR	Spurious Free Dynamic Range
SNDR	Signal to Noise and Distortion Ratio

SNR	Signal to Noise Ratio
SR	Slew Rate
THD	Total Harmonic Distortion

Chapter1. Introduction to ADCs

Analogue to digital converters are the most important building blocks in lots of applications. As electronics and telecommunication worlds are moving fast towards digitalization and there is an ever increasing demand on speed and accuracy of the processed data, the need for high speed and high resolution ADCs has grown dramatically over recent years.

1.1 Brief Review of ADC Architectures

Predominantly, ADC applications fall into four market categories [1]: 1) data acquisition, 2) precision industrial measurement, 3) voice band and audio and 4) high speed. Figure 1-1 shows the relation between these categories, resolution and speed with choice of ADC's architecture.

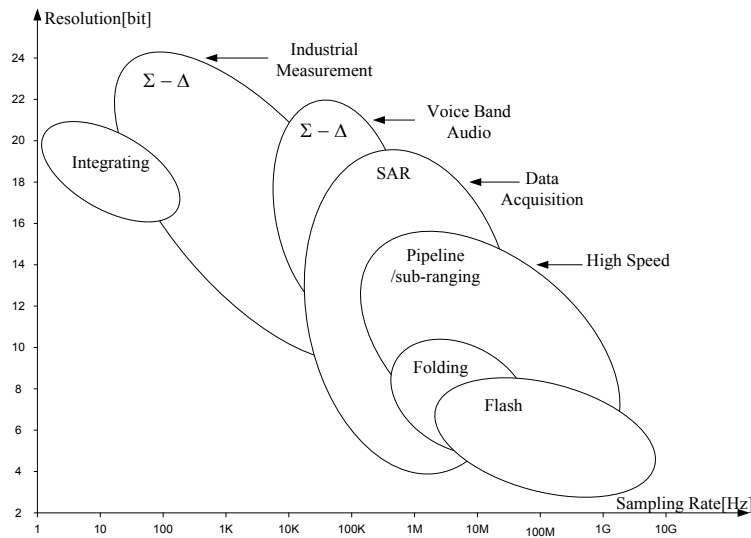


Figure 1-1: Speed and Resolution of Different ADCs [1]

Pipelined ADC is the architecture of choice in high speed and medium resolution applications. Examples of these applications are instrumentation, communications and consumer electronics.

The choice between different architectures can be made based on the speed, resolution, area and power consumption requirements in the target application. Knowing the specification, one can choose between different architectures to achieve the needed performance. Among available ADC architectures, flash, folding, sub-ranging and pipelined ADCs are fast enough to be considered as a high speed ADC. Below, ADC architectures are briefly reviewed.

1.1.1 Flash ADC

Flash ADCs are used in high speed applications. They convert the sampled data to digital output in one sample period, i.e. all bits are prepared in parallel and are available at the

output of the ADC at the same time. Due to inherent parallelism in flash architecture, the time needed for the result to be ready is equal to comparator's response time plus the time needed in decoder. The speed can be as high as tens of Giga hertz. Usually, the resolution of the flash ADCs is less than 8 bits. The architecture of a 2-bit flash ADC is illustrated in Figure 1-2 (a):

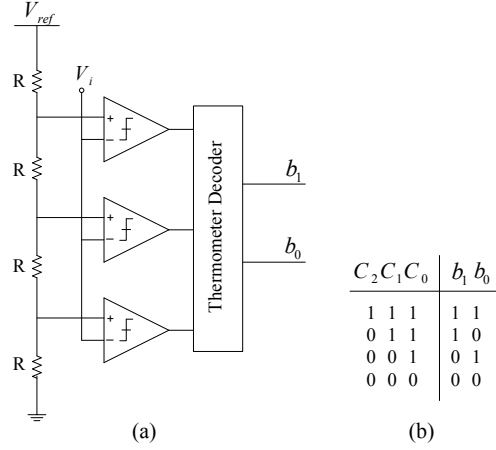


Figure 1-2: (a) 2-bit Flash ADC (b) Thermo-Code to Digital-Code Table

An N-bit flash ADC needs $(2^N - 1)$ reference voltages which are usually provided by a resistor ladder with 2^N identical resistors. Therefore, $(2^N - 1)$ comparators are needed to compare the input sample with the reference voltages in flash ADC. The result of this comparing is the generation of 3-bit thermometer codes as shown in the table of Figure 1-2 (b). A thermometer decoder is needed to convert these codes to binary. As can be seen, adding one bit to the resolution doubles the number of comparators needed which almost doubles ADC's power dissipation. An extra bit in resolution, also increases the accuracy requirements on comparators, therefore, more accurate reference voltages are needed. As a result, flash ADCs are not suitable for applications that need high resolution ADCs.

1.1.2 Folding ADC

Folding ADCs are categorised as multi-stage ADCs. The difference between a binary stage and a folding stage is that in folding ADC the output digital code is a Grey code and the residue signal resulted in each stage is a little bit different. Suppose that the input is a ramp between $0 - V_{ref}$ as in Figure 1-3 (a), the residue signal for a binary stage is shown in Figure 1-3 (b). When input signal is less than $\frac{1}{2}V_{ref}$ residue signal increases from $0 - V_{ref}$ and when input signal crosses $\frac{1}{2}V_{ref}$ the residue signal experiences a discontinuity and starts from 0 again. But, in a folding stage (Figure 1-3 (c)) there is no discontinuity and the residue signal starts to decrease from $V_{ref} - 0$. The mitigation of these discontinuities allows the converter to operate faster than binary implementation.

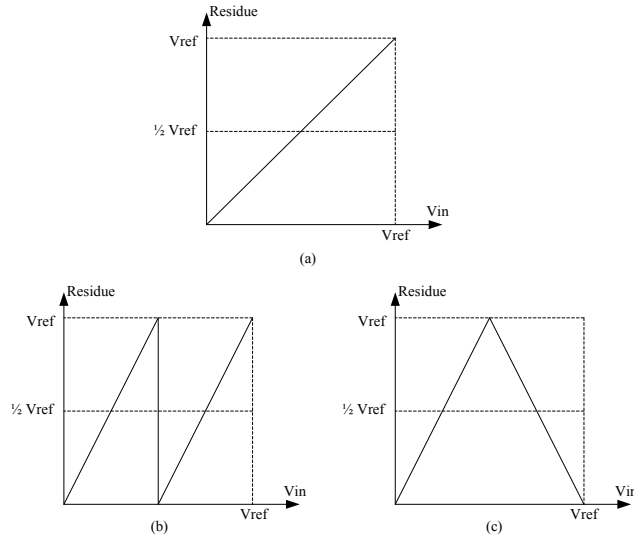


Figure 1-3: (a) A Ramp Input Signal, (b) Residue from a Binary Stage, (c) Residue from a Folding Stage

In Figure 1-4 the concept of the folding stage is illustrated [2]. The input signal is sampled and compared against $\frac{1}{2}V_{ref}$. The result is one bit grey code as the digital output of the stage. Based on the comparison, the switch position is decided. Pos1 is for inputs less than $\frac{1}{2}V_{ref}$ and Pos2 for inputs larger than $\frac{1}{2}V_{ref}$. The residue signal is shown in Figure 1-3 (c).

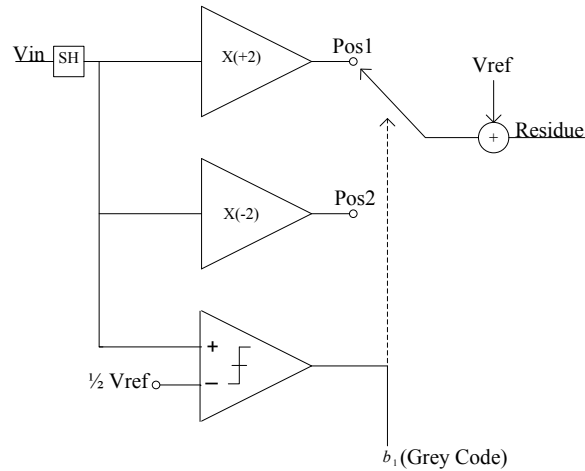


Figure 1-4: Concept of a Folding Stage

Using the folding stage in multi-stage architecture forms a folding ADC. Similar to other multi-stage architectures, this ADC also needs time alignment and the digital output can be digitally corrected. It is trivial to remember that somewhere, after digital outputs were aligned, there is a need for Grey code to binary code converter if the digital outputs are going to be used in a binary system after ADC, which is usually the case.

Folding ADCs have high speed conversion rates. The sampling frequency can be as high as a few hundred mega hertz. They can be used in applications that need medium resolution ADCs.

1.1.3 Sub-Ranging ADC

The idea behind sub-ranging ADCs is to use low resolution high speed sub-ADCs in a multi-stage design. Usually, sub-ranging ADCs are limited to 2 stages and they can be resolve up to 8 bits without any kind of digital correction scheme [1]. Pipelined ADCs' architecture stems from this architecture. A 6-bit two-stage sub-ranging ADC is illustrated in Figure 1-5:

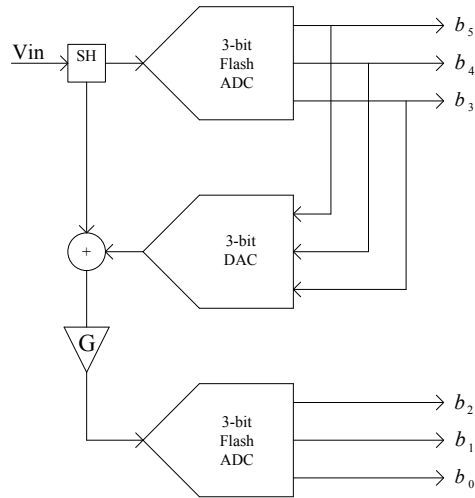


Figure 1-5: 6-bit Sub-Ranging ADC

In this ADC input voltage is sampled and converted into digital by a low resolution Sub-ADC (3 bits in this example) which resolves the upper three bits of the digital output. The bits resolved are converted back to analogue by the 3-bit DAC. The analogue output of the DAC is subtracted from the sampled input and the result is a residue signal which is amplified within the range of the next 3-bit Sub-ADC. The residue signal is converted to digital to form the lower three bits of digital output. Two-stage architecture results in latency in the time of data conversion completion, but the data conversion rate is one conversion per sampling period.

Sub-ranging ADCs can be more than two stages and resolve more than 8 bits, but this necessitates time alignment and digital correction. The concept of time alignment and digital correction is explained Chapter2 for pipelined ADCs.

1.1.4 SAR ADC

SAR ADCs are suitable for applications with the need of medium to high resolution (8-16 bits) and sample rates less than 5MS/s. They also consume low power which makes them right architecture for low-power applications. The principle behind a SAR ADC is shown in Figure 1-6:

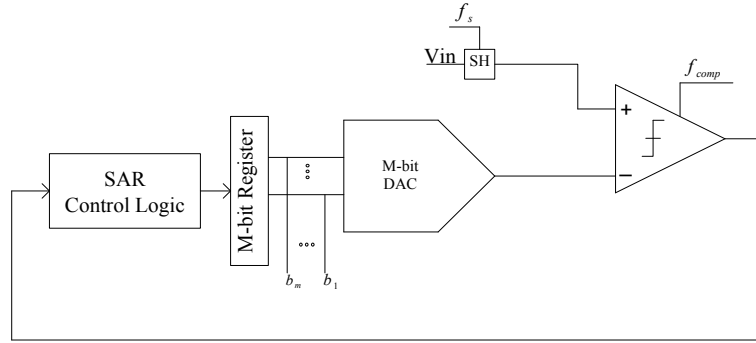


Figure 1-6: SAR ADC

Analogue input is sampled and held by the sample and hold circuitry. The sample is compared with the DAC's output and the decision is used in SAR control unit to set one bit digital resolved per each comparison (from MSB to LSB) and set the register to initial next digital to analogue conversion.

At the very beginning of conversion, register is set to digital value of $\frac{1}{2} V_{ref}$ (which is 100 for a 3-bit ADC) and after digital to analogue conversion, this value is compared with sampled data. If the comparison result would be a 1, the control unit keeps the MSB 1; else it forces the MSB to zero. Then the control logic sets next bit to one and the DAC function and comparison take place afterwards. This repetitive action goes on until all of the bits in register have been decided for. It is obvious that for an N-bit SAR ADC N comparison period is needed and only after that a new sample can enter the ADC to be converted to digital. Therefore, the SAR ADC's speed is limited to setting time of DAC, comparator's speed and the logic overhead [3].

1.1.5 Σ - Δ ADC

Σ - Δ ADC is mostly famous because of its noise shaping characteristics which results in higher SNR [1]. The noise shaping characteristics plus digital filtering and decimation moves most of the quantization noise to the outside of the Nyquist bandwidth and removes the out of band noise. As can be seen in the Figure 1-7, the input signal enters an ADC cell with oversampling ratio of K. After data conversion and noise shaping, the noise is filtered by a digital filter and the output rate is reduced to the sampling rate by a decimator. For each doubling of the oversampling ratio, the SNR within the Nyquist bandwidth ($f_s/2$) is improved by 3dB.

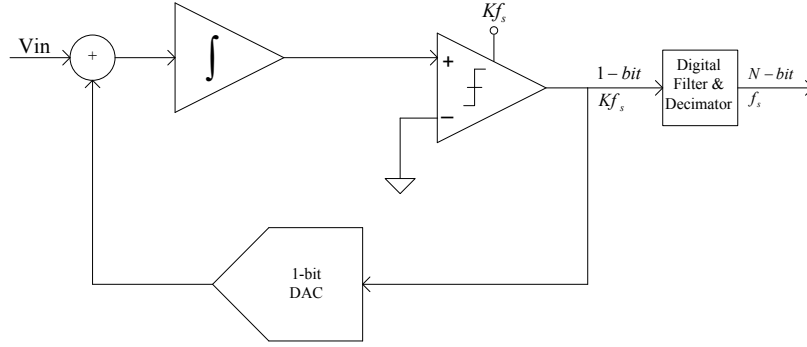


Figure 1-7: Σ - Δ ADC

As the ADC's resolution increases, noise shaping in Σ - Δ ADC becomes less effective. To increase the power of noise shaping, another level of integration can be added to the circuit which results in more complex circuitry. Another solution is to use multi bit architecture instead of 1-bit Σ - Δ modulator [1].

Σ - Δ ADCs can have resolutions up to 24 bits but their speed is limited to a few hundred hertz.

1.2 ADC Error Sources and Performance Metrics

Error in reference voltages due to manufacturing process will introduce error to the gain and offset of the ADC's transfer function. From the circuit implementation point of view, the main error sources in a pipelined ADC are gain, offset and nonlinearity errors in the sub-ADC and MDAC. Gain, offset and nonlinearity errors of the sub-ADCs in all stages, except for the last stage, can be corrected by the redundancy and digital error correction logic [4]. Last stage's errors are scaled down by the combined inter-stage gain of all preceding stages. Some of the offset error of the DAC can be corrected by digital correction; some is referred to the input of the ADC as an extra offset that can be cancelled by adding offset to the input. However, the requirement on the linearity of the DAC is high, especially for early stages.

Another error in an ADC is the quantization error. Quantization error is due to quantizing a continuous signal into discrete values [5]. This error can be treated as a white noise, especially when the resolution of the ADC is high (larger number of quantization steps in the transfer function). Ideally, the quantization noise is less than one quantization step which is equal to one LSB. The power of this noise can be calculated as in Equation 1-1 [6]. Where Q stands for quantization step and E for quantization error.

$$\text{Equation 1-1: } \overline{P}_q^2 = \frac{1}{Q} \int_{-Q/2}^{+Q/2} E^2 dE = \frac{Q^2}{12}$$

The ratio between the full-scale input signal's power and this noise power leads to the famous formula $SNR = 6.02N + 1.76$ for an ideal ADC. Quantization noise increases the noise floor of the ADC.

In order to verify ADC's performance and be able to compare different ADCs, a number of performance metrics are defined [5], [7], and [8]. These metrics are categorised into two groups, static performance metrics and dynamic performance metrics.

1.2.1 Static Performance Metrics

As mentioned above as a result of limited manufacturing accuracy some of reference voltages may slightly differentiate from the exact designed value, introducing gain and offset errors to the ADC's transfer function. The metrics to quantify ADC's static performance are:

- Integral non-linearity (INL): The maximum absolute value of differences between the ideal and actual code transition levels after correcting for gain and offset
- Differential non-linearity (DNL): The maximum absolute value of differences between the actual code widths and ideal code width (1xLSB)

In an ideal ADC, INL error is at most $\frac{1}{2}$ LSB and DNL error is 0xLSB, which is not the case in actual ADCs. The concept of INL and DNL is shown in Figure 1-8.

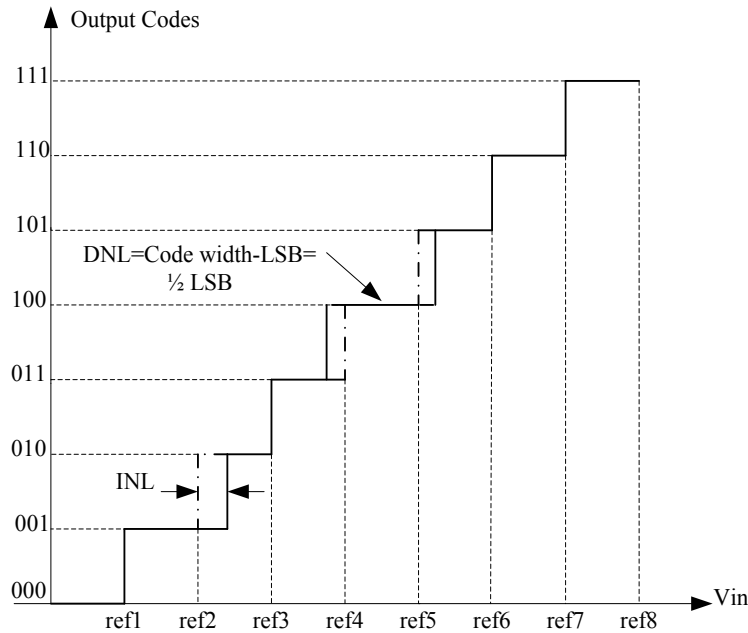


Figure 1-8: INL/DNL Concept

As can be seen in figure above, voltage references 2, 4 and 5 have deviated from their ideal value, producing non-linearity to the transfer function of a 3-bit ADC. The input voltage is assumed to be a ramp signal.

1.2.2 Dynamic Performance Metrics

Dynamic performance of the ADC is its performance regarding input signal and sampling frequency. To measure ADCs performance, a number of metrics are defined [7].

- **Signal to Noise ratio (SNR):** The ratio of the root mean square (rms) value of the signal power (S_p) to the noise power (N_p) at the output of the ADC, measured when applying a sinusoid, typically expressed in dB:

$$SNR = 20 \log\left(\frac{S_p}{N_p}\right), dB$$

- **Spurious Free Dynamic Range (SFDR):** The ratio of the rms value of the signal power (S_p) to the rms value of the largest spur power (P_{spur}) at the output of the ADC, measured when applying a sinusoid, typically expressed in dB:

$$SFDR = 20 \log\left(\frac{S_p}{P_{spur}}\right), dB$$

- **Total Harmonic Distortion (THD):** The ratio of the rms value of the signal power (S_p) to the mean value of the root-sum-square of all harmonics' power (D_p) at the output of the ADC, measured when applying a sinusoid, typically expressed in dB:

$$THD = 20 \log\left(\frac{S_p}{D_p}\right), dB$$

- **Signal to Noise and Distortion ratio (SNDR/SINAD):** The ratio of the rms value of the signal power (S_p) to the mean value of the root-sum-square of the all harmonics' power plus noise components ($N_p + D_p$) within the Nyquist bandwidth at the output of the ADC, measured when applying a sinusoid, typically expressed in dB:

$$SNDR = 20 \log\left(\frac{S_p}{N_p + D_p}\right), dB$$

- **Effective Number of bits (ENOB):** The actual resolution of the ADC in presence of noise and distortion, when applying a full scale input signal, extracting N from SNR equation for an N-bit ideal ADC ($SNR = 6.02N + 1.76$) and substituting SNR with SNDR:

$$ENOB = \frac{SNDR - 1.76}{6.06}$$

Chapter2. Pipelined ADC

Pipelined ADC is built from several low resolution converters in a pipeline. The number of stages and the number of bits resolved by each stage along with redundancy bit(s) should be determined wisely considering power, speed and resolution of the ADC and accuracy requirements on sub converters. Most of the time, in high speed ADCs lower resolution per stage is chosen to have lower inter-stage gain and settling time which results in higher conversion rate. Low resolution per stage also relaxes the requirement on accuracy of voltage references in Sub ADC and comparators. Drawbacks of having lower bits resolved in stages are higher number of stages that are needed and more noise and gain and offset errors from latter stages brought back to the input due to lower inter-stage gain and will lower the total ADC's accuracy. Usually in high resolution ADCs, more bits are resolved in each stage. Higher resolution per stage gives the benefit of having higher inter-stage gain which will reduce the later stages' noise contribution to the overall noise of the ADC. However, this increases the power dissipation of the ADC and also the area required for the ADC. The noise and other errors of subsequent stages are reduced by former stages' squared gain. Adding more bits to be resolved in early stages, especially stage1, will relaxes the requirements on following stages' accuracy and noise requirements and will allow scaling to be applied to them. This technique helps with area and power limitations.

Stages can also have redundancy bit that can be shared between neighbouring stages by overlapping. This technique leaves room for error correction (does not produces 111) and adds $\frac{1}{2}$ LSB offset to prevent saturation of coming stages due to comparison errors occurred in present stage. This offset helps to keep the residue signal within the 0-Vref range of the ADC. In Figure 2-1, it can be seen that even very small deviations from the ideal value in reference voltages produces a residue voltage larger than V_{dd} or lower than V_{ss} . This out of bound voltage will saturate next stages. Another advantage of this technique is the reduced inter-stage gain for higher number of resolved bits. For example in a 2.5 b stage with 3 raw bits and 2 resolved bits (one redundant bit) from total bits of the ADC, stage gain will be 2^2 instead of 2^3 . Reduced gain will relax the requirements on the OpAmp employed in the MDAC. Redundant bit can be added to any sub ADC with different resolution.

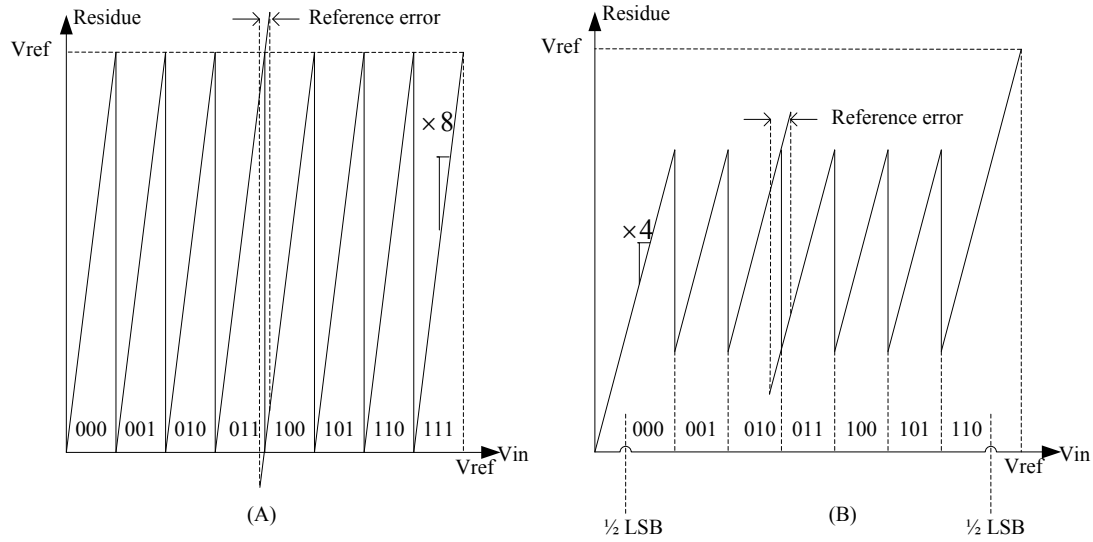


Figure 2-1: Error Caused by Reference Voltage Deviations from Ideal Value in (A) 3-bit Stage and (B) 2.5-bit Stage

2.1 Pipelined ADC's Architecture

A 12-bit pipelined ADC incorporating 2.5 b stages is shown in Figure 2-2:

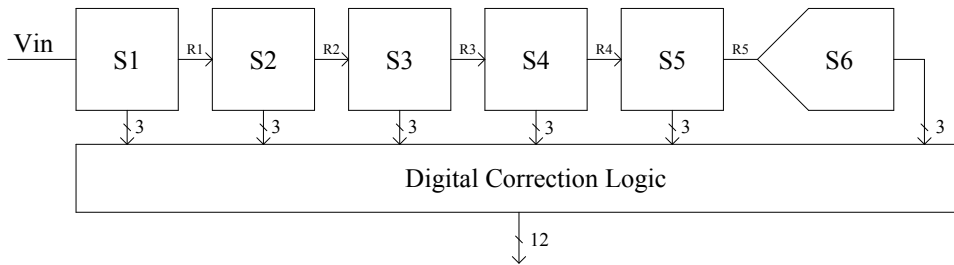


Figure 2-2: 12-bit Pipelined ADC

The ADC incorporates 6 stages; each one (except for stage 6) consists of a sample and hold, DAC, subtraction and amplification circuitry (all of which known as multiplying DAC or MDAC) and a low resolution but high speed flash ADC. Stage 6 is a 3-bit flash ADC.

In Figure 2-3 one stage of pipelined ADC is represented:

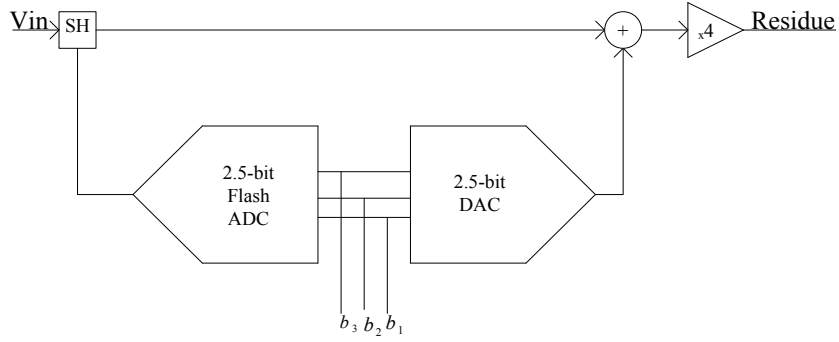


Figure 2-3: Pipeline Stage

Inside each stage input voltage is converted to 3 raw bits by the high speed flash ADC and then reconstructed back to analogue by the DAC. The reconstructed signal is subtracted from original sampled signal and the difference is multiplied by the amplification factor, producing the residue signal. The residue signal is applied to the next stage to be processed and the current stage starts sampling the incoming signal and processing on the sampled and held data. The pipelining operation produces latency to the digital data production but after that there will be one conversion per clock cycle. As a result of this concurrency conversion rate of the ADC is independent of the number of stages. The residue signal is shown in Figure 2-4:

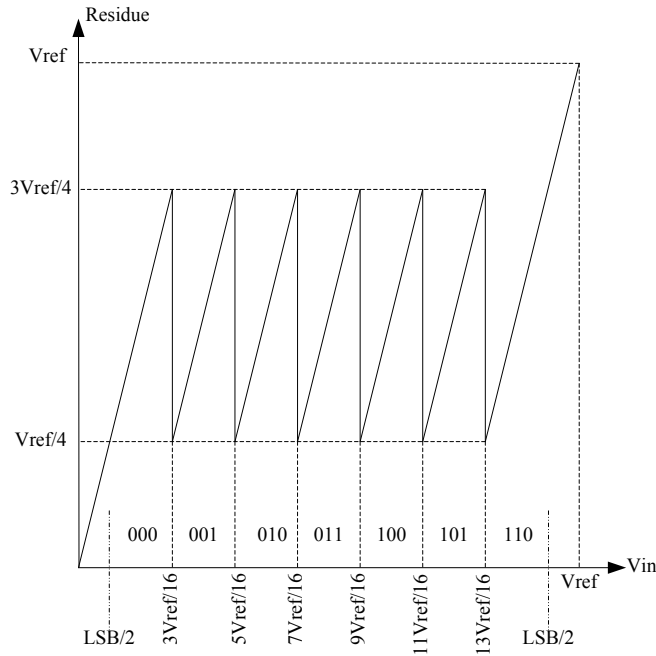


Figure 2-4: Residue Signal of A 2.5b Stage

Reference voltages for 2.5b flash ADC to be used in comparators are $\frac{3}{16}V_{ref}$, $\frac{5}{16}V_{ref}$, $\frac{7}{16}V_{ref}$, $\frac{9}{16}V_{ref}$, $\frac{11}{16}V_{ref}$ and $\frac{13}{16}V_{ref}$. These references are applied to six comparators of the

flash ADC along with the sampled and held signal. The correction range of the ADC is $\frac{1}{4}V_{ref}$. In case gain and offset errors occur, as long as the error stays within this range, it can be corrected by digital correction and coming stages will not be saturated.

2.2 Flash Sub-ADC

Designed pipelined ADC has fully differential architecture. Fully differential architecture allows more dynamic range and reduces even harmonics' effect on nonlinearity. One out of six segment of the sub-ADC is presented in Figure 2-5 [10]:

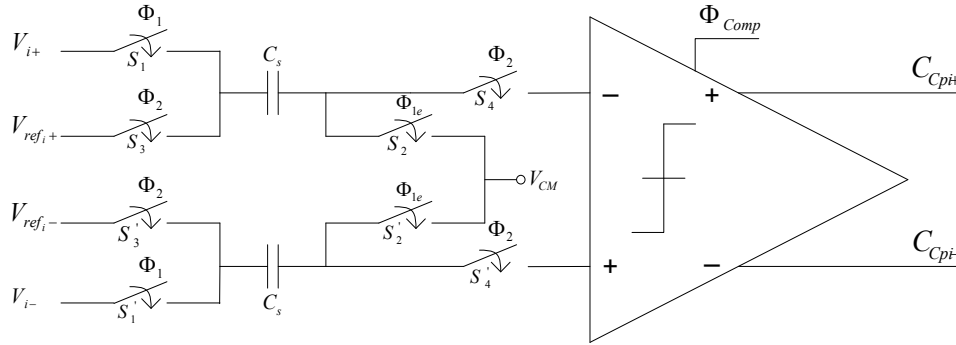


Figure 2-5: One Segment of Comparing Circuitry in Sub-ADC

Each sub-ADC includes six segments shown in figure above. Input signal is sampled during phase1 into C_s when switches S_1 and S_2 (S_1' and S_2') are closed (Figure 2-6 a). S_2 (S_2') turns off before S_1 (S_1'), injecting charge into C_s [11]. This charge ($\Delta q_2 = W_2 L_2 C_{ox} (V_{gs2} - V_{th2})$) appears as an offset voltage added to the sampling capacitor's voltage. Fully differential architecture mitigates this offset voltage and it will have no effect on the output voltage. The sampling period is determined by clock1e. Switch S_1 (S_1') opens after S_2 (S_2') and switches S_3 (S_3') and S_4 (S_4') turn on after S_1 (S_1') turned off. Since left plate of sampling capacitor (C_s) was connected to V_{i0} at the moment when S_1 (S_1') turned off and is connected to V_{refi} when S_3 (S_3') turns on (two constant voltages), the charge injection and charge absorption by switches S_1 (S_1') and S_3 (S_3') will not introduce an error to the final value.

Sampled voltage is held during phase2 (Figure 2-6 b):

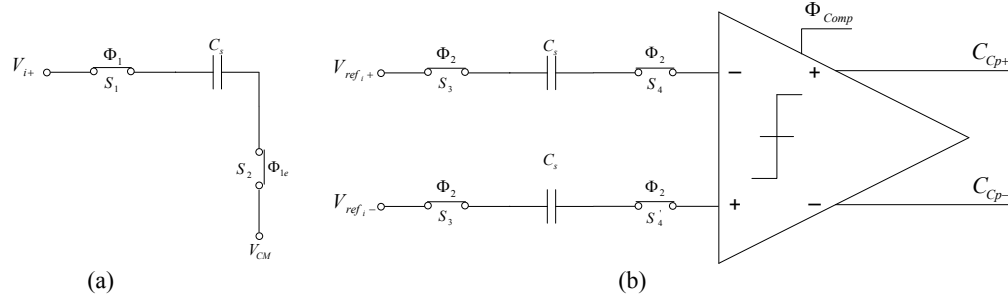


Figure 2-6: (a) Sampling Phase in Flash Sub-ADC, (b) Comparing Phase in Flash Sub-ADC

The sampled data is compared against six reference voltages V_{ref_i} ($\frac{3}{16}V_{ref}$, $\frac{5}{16}V_{ref}$, $\frac{7}{16}V_{ref}$, $\frac{9}{16}V_{ref}$, $\frac{11}{16}V_{ref}$ and $\frac{13}{16}V_{ref}$). The result from this comparison gives six differential pairs of thermometer codes (C_{cp1-6+} , C_{cp1-6-}). After producing these codes, they have to be converted to 3 bits binary codes.

Comparators clock is delayed version of clock2. In comparator's circuit pre-amplification is used to amplify small differences between input and reference voltage to increase the accuracy of the comparator. The pre-amp circuit needs time to settle and the delay allows the output to reach its final value to be used in comparison.

2.2.1 Thermometer Decoder

The thermometer decoder can be implemented using lots of techniques, for example by using pass-transistors, multiplexing, etc. In this design thermometer codes are used as address bits of an OR-based ROM. Figure 2-7 shows a 3-to-2 bit thermometer to binary decoder (Figure 1-2-b), using the ROM implementation. The address decoder circuit is OR-based designed as well. All address and data lines in the address decoder and ROM are connected to V_{dd} through PMOS devices which are always on. Whenever a line in the ROM should be chosen, all transistors in that line should be turned on which means the address line should be kept high. For an address line to be high, all transistors that are connected to it should be off. For example, if $C_2C_1C_0$ is 000 ($V_{in} \leq V_{ref1}$) then Add1 is V_{dd} and the transistors in the first line turn on, bringing data lines to 00 which is the binary output expected for $V_{in} \leq V_{ref1}$.

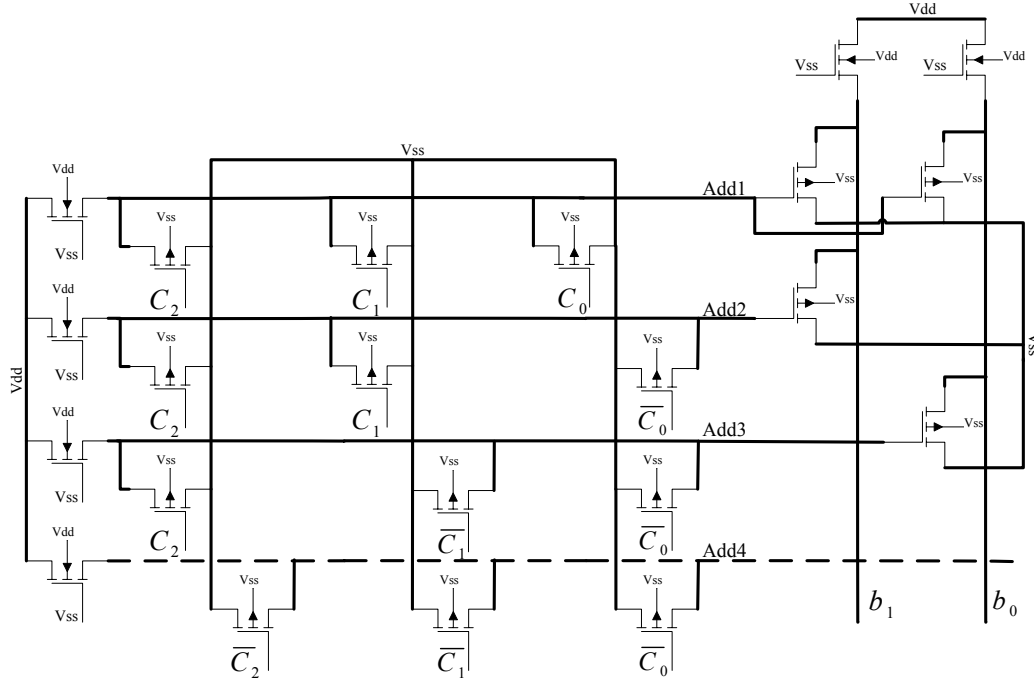


Figure 2-7: Thermometer to Binary Decoder Implemented by OR-Based ROM

In picture above, the last address line (dashed line) is not needed to be implemented, as it does not drive any transistor in the ROM. It has been kept in the picture for the sake of more accuracy. The actual design is fully differential 6-to-3 bit decoder (2.5bit/s implementation).

2.2.2 Comparator

Comparators are made of two basic building blocks, a preamplifier and a latch. The comparator is used to resolve small input signal and produce a digital 0 or 1 output. Therefore, the amplifier does not have a linearity requirement. It should amplify the small input signal enough to make the latch change its state if necessary. The basic concept of a comparator is shown in Figure 2-8:

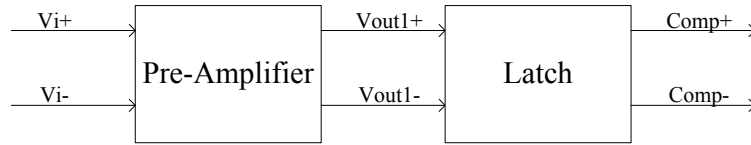


Figure 2-8: Basic Concept of a Comparator

The comparator operates in two phase, reset and evaluation (latching). In reset phase, the latch is pre-charged to V_{dd} to reduce the power dissipation in this phase. In evaluation phase, the amplified input signal causes the latch to change its state in either direction and by the aid of positive feedback the output signal will clip to one of the supply sources, producing the digital outputs. The latch circuitry is depicted in Figure 2-9 [10]:

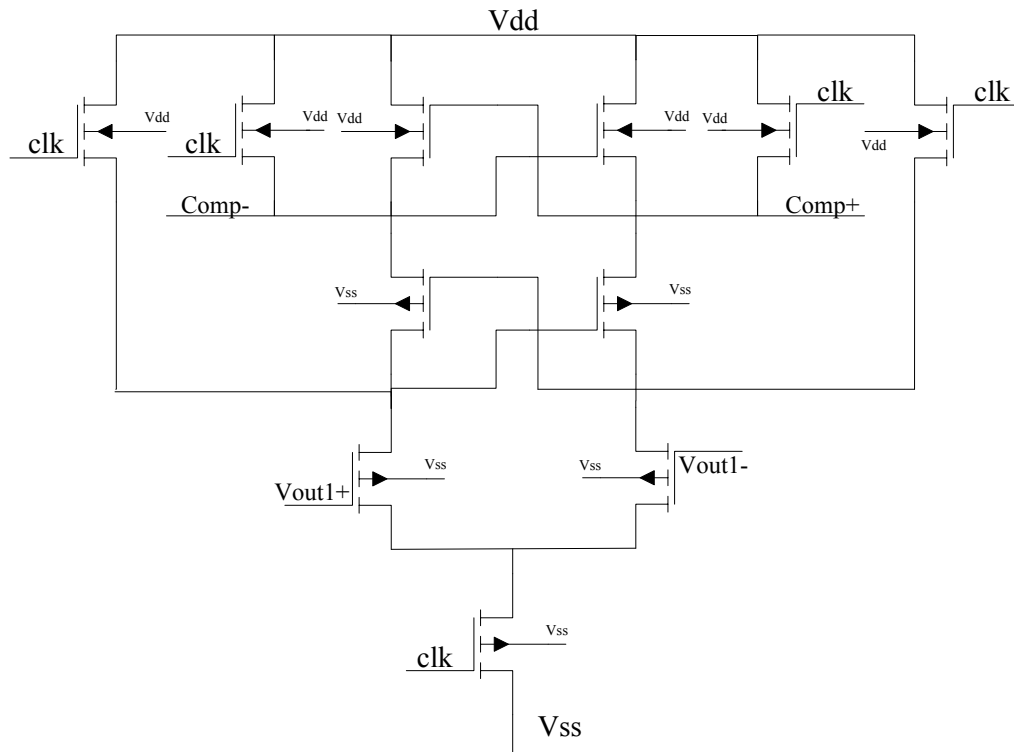
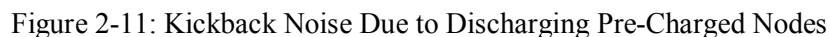


Figure 2-9: Latch Circuitry of The Comparator

Pre-amplifier in the comparator helps with very small input signals, i.e. when the difference between the sampled input signal entering the comparing circuitry of the sub-ADC and the reference voltages of the flash ADC is very small to cause a change in the state of latch. Pre-amplifier also prevents the kickback noise from flowing into the driving circuitry and suppresses noise and offset of the latch when referring to the input. The gain of the pre-amplifier is determined by the accuracy needed, but, it is usually between 4-10 dB. Choosing a gain more than 10 dB will reduce the speed of the comparator. Therefore, in high speed applications, the gain should be chosen more carefully. The pre-amplifier circuit, shown in Figure 2-10, is scaled down one stage non-boosting amplifier designed for the MDAC (studied in Chapter2).



When the latch goes from reset mode into evaluation mode, there is a charge transfer either into or out of the inputs of the latch. The charge which transfers from input to the circuit is the charge needed to turn on the transistors in positive feedback circuitry and the charge which flows back to the inputs is the charge that is needed to be removed from pre-charging transistors (Figure 2-9). Another charge that should be considered is the charge introduced to the circuit when discharging the pre-charged nodes of the circuit, nodes A and B in Figure 2-11, at the drain of input differential transistors. This charge is transferred to the input nodes by the gate-drain capacitor of input pairs. If node C, in figure below is pre-charged as well as nodes A and B, then the charge removed from this node also contributes in kickback noise through C_{gs} . As explained before, using pre-amplifier can eliminate this noise.



2.2.2.2 HYSTERESIS

When comparator changes its state, it has a tendency to stay in that state [12]. This tendency is called hysteresis and can be eliminated by pre-charging differential nodes or connecting or connecting differential nodes together, using switches, before entering evaluation mode.

2.2.2.3 METASTABILITY

When the comparator's output is neither a 1 nor a 0, the output is considered as meta-stable [13]. The problem can be reduced by allocating more time to latching process and/or using Grey encoding (which allows one transition at a time) and then Grey to binary decoding. A meta-stable output can be translated into a 1 or a 0 by the following circuit; so, in order to avoid detrimental errors, each comparator's output should drive one circuit at a time.

2.3 MDAC

An MDAC performs sampling, digital to analogue conversion, subtraction and amplification. The circuit shown in Figure 2-12 is responsible for sampling, subtraction and amplification in an MDAC:

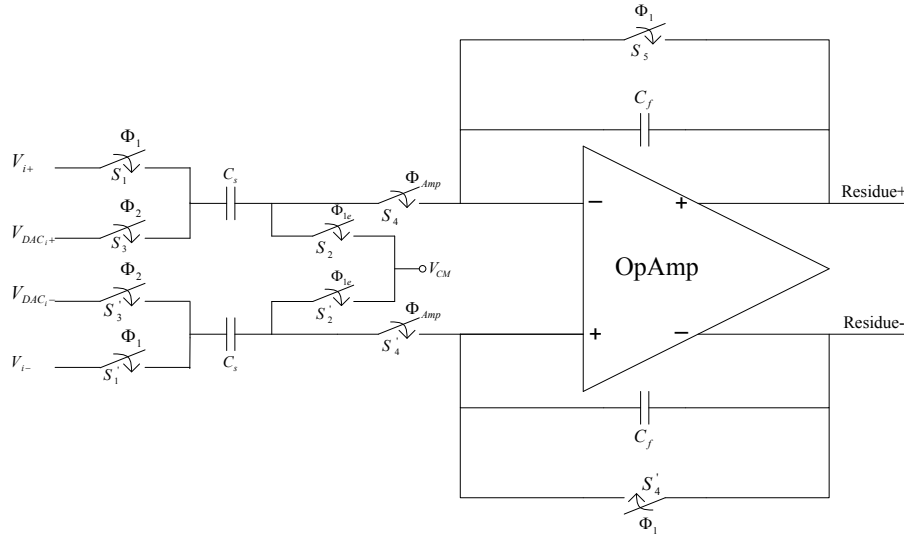


Figure 2-12: Sampling and Multiplication Part of The MDAC Circuit

Amplifier's clock is a delayed version of comparator's clock. This delay is needed for thermometer decoder and DAC to complete the conversions from thermometer codes to binary codes and from digital codes to analogue signal.

During phase1 input voltage is sampled into C_s when switches S_1 and S_2 (S_1' and S_2') are on (Figure 2-13). Like sub-ADC, S_2 (S_2') turns off before S_1 (S_1'), leaving node A (B)

float and introducing a constant offset to the sampled voltage (cancelled by differential implementation). The OpAmp is placed in the unity-gain feedback during this phase and output voltage resets to its common mode voltage.

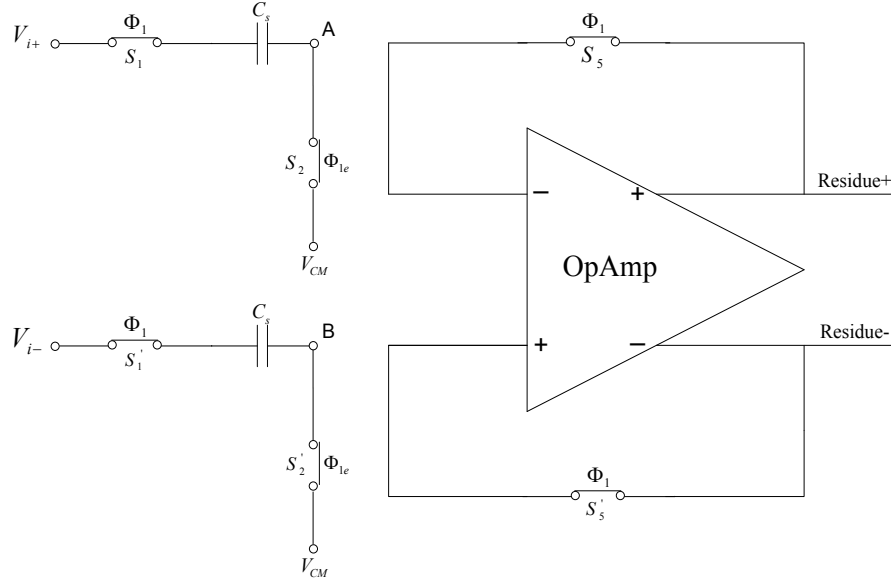


Figure 2-13: MDAC in Sampling Mode

The amplification mode is presented in Figure 2-14. As explained in sub-ADC section charge injection by switch S_1 (S_1') or absorption by switch S_3 (S_3') will not introduce an error to the final value. Switch S_5 (S_5') turns off before switch S_4 (S_4') adding a constant charge to the input node of the amplifier. This charge equals $\Delta q_5 = W_5 L_5 C_{ox} (V_{gs5} - V_{th5})$ which produces an error into the output [11]. Half of this charge goes directly to the output node, causing temporarily glitch. Another half flows back to the Input node of the OpAmp which is virtual ground, so, the charge is conserved at this node. Then, the charge resides on the left plate of C_f . This charge introduces an error equal to $\frac{\Delta q_5}{2C_f}$ to the output

voltage. To compensate for this error dummy switches are used (Figure 2-15). If dummy switch's size is chosen such that $L_d = L_5$ and $W_d = \frac{1}{2} W_5$ then the charge injected by S_5 into the input node of amplifier ($\frac{\Delta q_5}{2}$) will be absorbed by S_d and vice versa. Use of dummy switches also helps with clock feed through error.

Switch S_4 (S_4') produces some error when turning on or off. This error is constant and independent of the input like the case explained for switch S_5 (S_5') and can be compensated for if necessary.

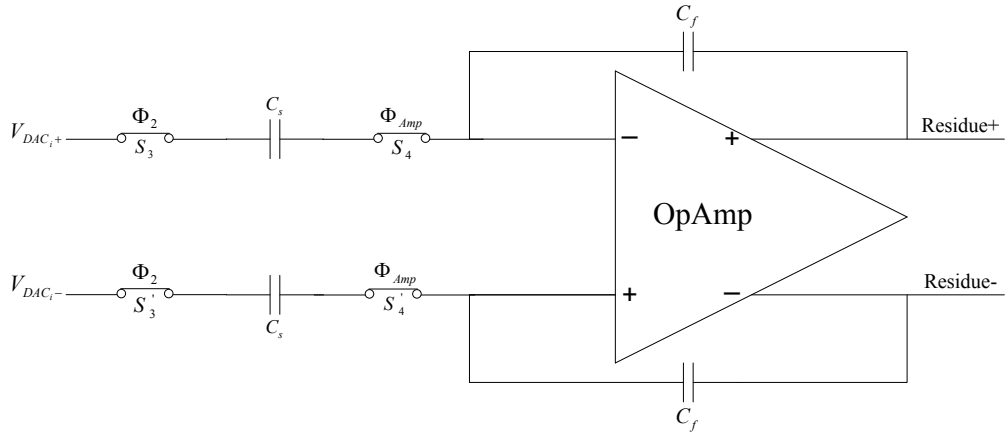


Figure 2-14: MDAC in Amplification Mode

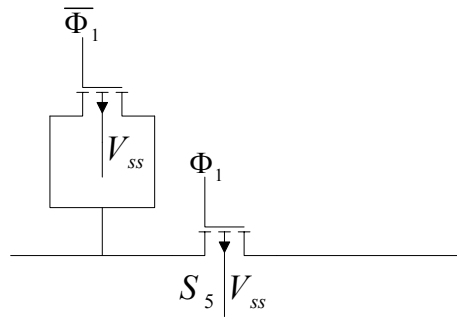


Figure 2-15: Use of Dummy Switches to Compensate for Charge Injection

2.3.1 Resistive Ladder DAC

DAC's transfer function versus input changes is shown in Figure 2-16. The DAC's reference voltages, for 2.5 bit architecture, are 0 , $\frac{1}{6} V_{\text{ref}}$, $\frac{2}{6} V_{\text{ref}}$, $\frac{3}{6} V_{\text{ref}}$, $\frac{4}{6} V_{\text{ref}}$, $\frac{5}{6} V_{\text{ref}}$ and V_{ref} .

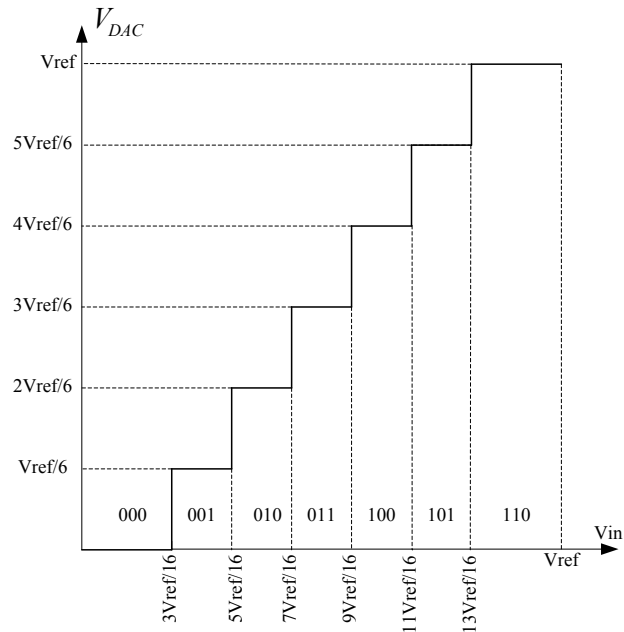


Figure 2-16: DAC's Transfer Function

A resistive ladder DAC is implemented and used in the stages. Switches involved with transferring high voltages are PMOS devices because of their better conductivity of high voltages. NMOS devices are used to conduct lower voltages.

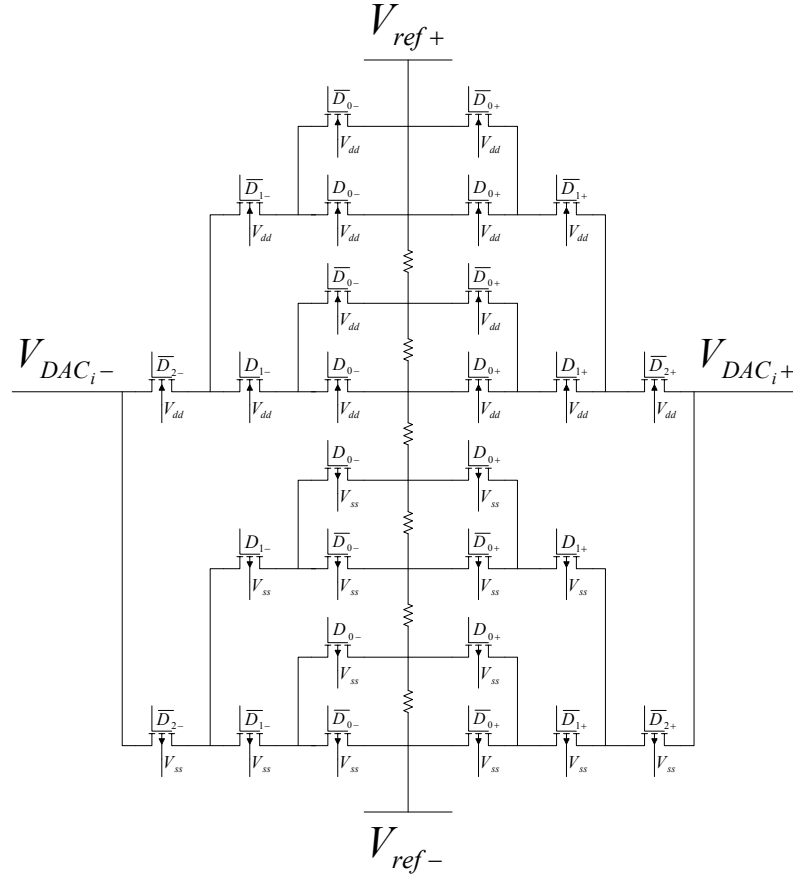


Figure 2-17: Resistive Ladder DAC

In implementation same ladder is used to provide flash sub-ADC and DAC reference voltages. In case of any mismatch and fabrication error, the reference voltages would have same errors and this will suppress nonlinearity.

2.4 Bootstrapping

High linearity requirement of the 12-bit ADC necessitates linear operation of the switches in the sub-ADC and MDAC structure. For a switch to work with high linearity, it should work with constant overdrive voltage. To serve this purpose some of the switches are bootstrapped, especially front end switches whose overdrive voltage suffers from the changes of input voltage. The bootstrap circuit, designed in [10] and adapted for low-voltage 65nm CMOS technology, is depicted in Figure 2-18:

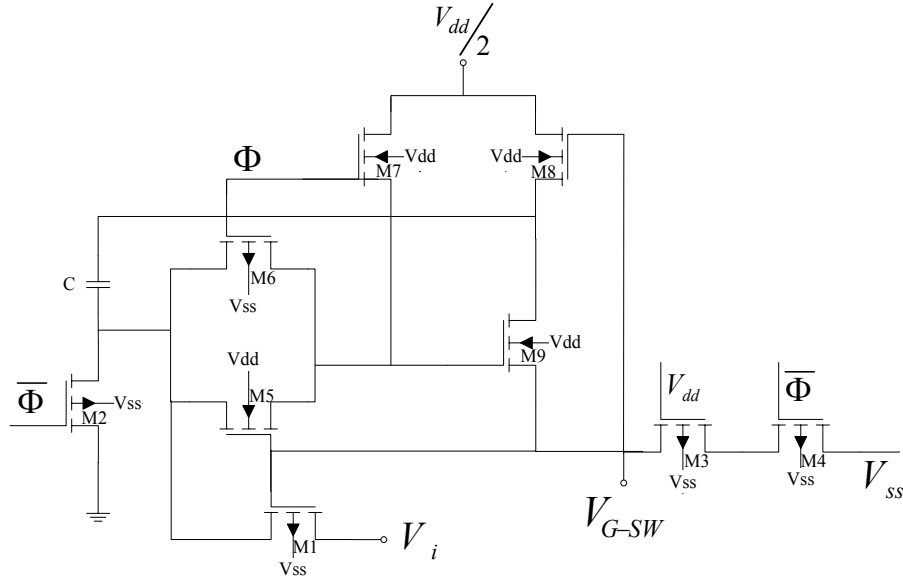


Figure 2-18: Bootstrap Circuit

When a clock signal is going to drive a switch, it can be applied to a bootstrap circuit and be manipulated to be more suitable as a driving gate voltage. In Figure 2-18, clock signal with phase Φ is applied to the bootstrap circuit and used to produce the signal V_{G-SW} which is the new driving gate voltage of the switch. When clock is low, $M_3 - M_4$ bring V_{G-SW} to V_{ss} and keep M_1 off and M_5 and M_8 on. Transistor M_3 is always on (gate voltage is V_{dd}) and used to shield output voltage from the switch M_4 's clock feed through. During this phase capacitor C will be charged to $\frac{1}{2} V_{dd}$ through switches M_2 and M_8 . Switch M_7 is also on during this phase and is responsible for keeping M_9 off by bringing gate voltage of M_9 to V_{dd} .

When the clock goes high, M_2, M_4 and M_7 turn off and M_6 turns on. At the very moment, as M_5 is on (V_{G-SW} is still zero), it conducts the bottom plate voltage of C (still zero as M_1 is off) to the gate of M_9 , turning it on and increasing V_{G-SW} to almost $\frac{1}{2} V_{dd}$. This voltage is enough to turn on M_1 switch and turning off M_5 and M_8 switches. Switch M_9 is bootstrapped itself as its gate-source connection is placed in parallel with C when M_6 is on (so, $V_{gs9} = \frac{1}{2} V_{dd}$). With a high clock voltage and through switches M_1, M_6 and M_9 , the output voltage becomes equal to $V_{in} + \frac{1}{2} V_{dd}$, which means that gate-source voltage of bootstrapped switch is now constant and independent of input voltage. This increases the linearity of the switch. Bootstrapping also helps with switches conducting constant high voltages. It can provide a high enough overdrive voltage for those switches.

2.5 Clocking Scheme

Clock phases needed within the stage are depicted in Figure 2-19:

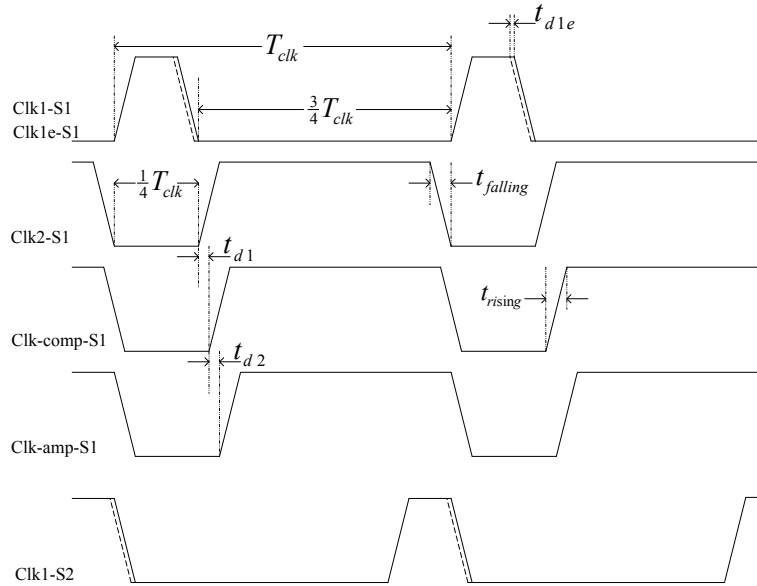


Figure 2-19: Stage Clock Phases

Clock1 is used to sample the input data by the sampling network in flash sub-ADC and MDAC simultaneously. Pulse width of this clock is almost $\frac{1}{4}$ of the sampling period. Clock1e is similar to clock1 in regards to period and 25% pulse width, but it turns off before clock1 to cancel charge injection problem from sampling switches. Allocating less time to sampling allows the circuit to spend more time on amplifying which gives amplifier more time to settle, increasing maximum sampling frequency.

Clock2 is used for introducing reference voltages to the sampling network to be compared to sampled data (in sub-ADC) or subtracted from it (in MDAC). The pulse width of clock2 is almost $\frac{3}{4}$ of sampling period. As explained before, in sub-ADC and MDAC sections, the comparators' clock and amplifier's are delayed version of clock2. All clocks' pulse width is lowered by rising and falling time to obtain non-overlapping clocks.

Sampling in each stage (except for stage1) starts at the last 25% of the amplification clock of preceding stage. This way, as the OpAmp amplifies the residue signal and resides within the accepted error ($\frac{1}{2}$ LSB) of its final value, the sampling capacitance of succeeding stage is charged with the residue signal to reach the final value simultaneously. Using this scheme, conventional sampling period can be reduced by 25%.

2.6 Digital Correction and Time Alignment

The bits from each stage are not resolved at the same time. As a result the output bits from 6 different stages that correspond to the same input sample are ready at different point in

time and should be time aligned and then digitally corrected. In order to align the bits related to the same sample shift registers are used (

Figure 2-20). The number of DFF in each shift register to pick the correct data is determined by clocking scheme and verified by simulation.

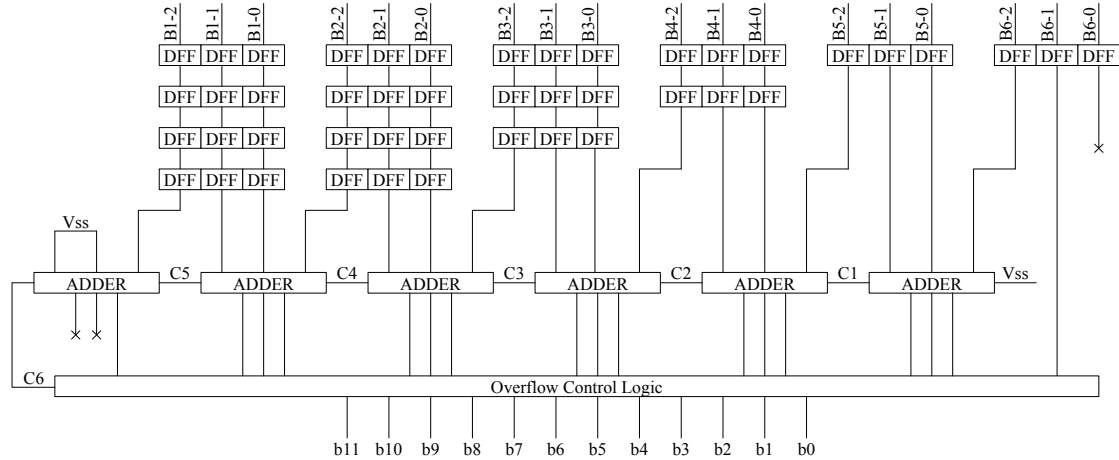


Figure 2-20: Time Alignment and Digital Correction Logic

The output bits from 6 stages, after being aligned, enter digital correction process which uses digital adders to overlap one bit from each stage with one bit from its neighbours. The procedure is shown in Figure 2-21:

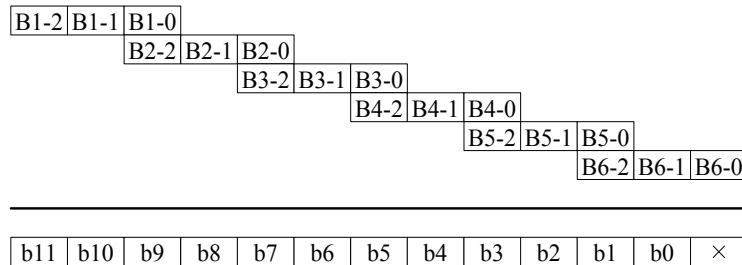


Figure 2-21: Digital Correction Logic

One bit redundant that was added to each stage allows the digital correction action which will help to correct for errors due to comparators offset and amplifiers gain error.

2.7 Noise Budgeting

When designing an ADC the tolerable noise within the system should be determined regarding the needed SNR and system's accuracy requirements. Calculated noise power will set the requirements on capacitor sizing. The output node of stage one is a suitable start point for this calculation [9]. The required SNR at this node can be calculated using Equation 2-1:

Equation 2-1: $SNR = 6.02 \times (N - M - 1) = 60.2dB$

Where N is the ADC resolution and M is the number of raw bits per stage (one bit redundant). Converting the dB value to the voltage measure gives $SNR \approx 1.3 \times 10^6$. Signal power at the output node of the stage1 equals:

Equation 2-2: $S_p^2 = \frac{(V_{FS}/2)^2}{2} = \frac{(700mv)^2}{2} = 0.245v^2 (rms)$

V_{FS} is the full scale differential input voltage to the second stage (in this case all stages). Therefore the allowable noise power at the output node of stage one is calculated as:

Equation 2-3: $S_n^2 = \frac{S_p^2}{SNR} = 0.19 \times 10^{-6} v^2 (rms) = 434.12 \mu v$

There are a number of contributors to the noise voltage at the output node of stages including jitter noise, active circuit noise. But the main contributor is the thermal noise, for which 50% of the allowable noise is reserved (around 218 μv). Therefore, the sampling capacitors' size for different stages should be calculated. Noise from sampling capacitors in first stage is the most important to be kept as low as possible. This is because noise at the output of other stages is divided by the cumulative squared inter-stage gain of preceding stages. Having more bits resolved in the first stage gives the benefit of lower input referred noise by introducing larger inter-stage gain.

In order to calculate the noise voltage, first of all the effective capacitor value at the output node of stage1, when the MDAC1 is in its amplification mode and stage2 has started to sample data, should be determined. The effective capacitor is determined in Equation 2-4:

Equation 2-4: $C_{eff} = 6 \times C_{comp2} + C_{s2} + \frac{C_{f1}(C_{s1} + C_{in1})}{C_{f1} + C_{s1} + C_{in1}}$

In equation above, third term $(\frac{C_{f1}(C_{s1} + C_{in1})}{C_{f1} + C_{s1} + C_{in1}})$ is the effective capacitance due to loading effect in a feedback system. C_{s2} and C_{comp2} are sampling and feedback capacitor of stage2, respectively. In this design same stage architecture is used, so, sampling and feedback capacitors are the same in all stages.

The calculated effective capacitor noise contribution can be obtained from Equation 2-5:

Equation 2-5: $V_n = \sqrt{\frac{2KT}{C_{eff}}}$

Where: $K = 1.38 \times 10^{-23} \text{ J/K}$; $T = 300^\circ \text{K}$; The $\times 2$ factor in the noise voltage equation stems from differential implementation and additive nature of thermal noise. Choosing $C_s = 800 \text{ f}$, gives thermal noise voltage less than $\frac{1}{2}$ LSB (LSB of a 12-bit ADC with 1.4v differential full-scale input amplitude), much less than allowable noise voltage at this node.

Chapter3. Introduction to the Fundamentals of OpAmps

Operational amplifiers (OpAmps) are basic building blocks of a wide range of analogue and mixed signal systems. Basically, OpAmps are voltage amplifiers being used for achieving high gain by applying differential inputs. The gain is typically between 50 to 60 decibels. This means that even very small voltage difference between the input terminals drives the output voltage to the supply voltage. In the case of using 65nm CMOS technology, this small voltage difference can be around tens of millivolts. As new generations of CMOS technology tend to have shorter transistor channel length and scaled down supply voltage, the design of OpAmps stays a challenge for designers.

In this chapter, an ideal OpAmp will be introduced. Later the performance parameters of OpAmp will be defined and then OpAmp's imperfections, which stems from the trade-offs between the parameters, will be briefly discussed. Following a review of simple topologies of OpAmp and a comparison between them, two techniques for achieving higher gain and output swing will be described. For the sake of simplicity, the OpAmp model used in this chapter is a single-ended operational amplifier with differential inputs as pictured in Figure 3-1.

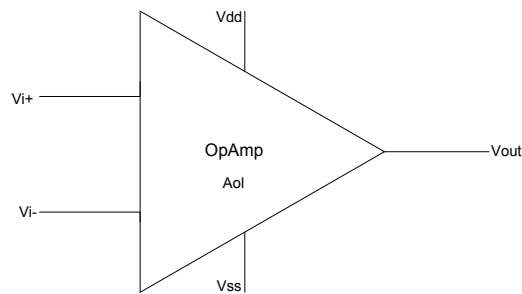


Figure 3-1 : A Single-Ended OpAmp Symbol

Where the notations on the symbol stand for:

- A_{ol} : Open- loop Gain
- V_{i+} : Non-inverting input
- V_{i-} : Inverting input
- V_{dd} : Positive Supply Voltage
- V_{ss} : Negative supply voltage
- V_{out} : Output

3.1 Ideal OpAmp

In the past, most OpAmps were designed to be used in many applications. This means that they have been designed as general purpose building blocks. This property leads to the idea of an ideal OpAmp with very high gain, high input impedance and low output impedance

disregard for the signal applied to the input. The transparent symbol for an ideal OpAmp is shown in the Figure 3-2.

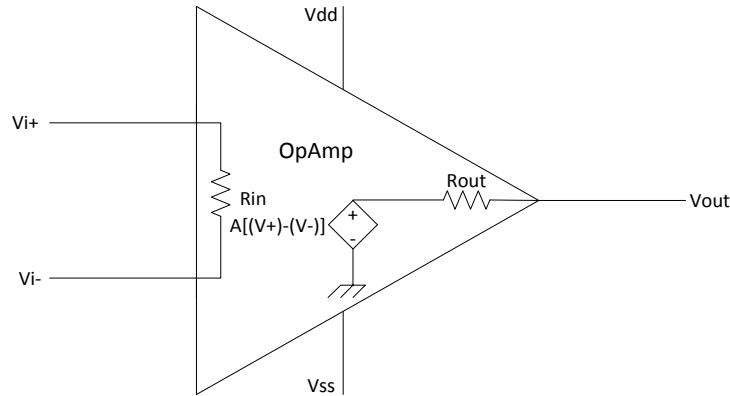


Figure 3-2: Ideal OpAmp

The Ideal OpAmp's properties:

- Infinite open-loop gain, $A_{ol} = \infty$
- Infinite input impedance, $R_{in} = \infty$
- Zero output impedance, $R_{out} = 0$
- Zero input current
- Zero input offset voltage (i.e., if $V_{i+} = V_{i-} \Rightarrow V_{out} = 0$)
- Infinite slew rate
- Zero noise.
- Infinite bandwidth
- Infinite Common-mode rejection ratio (CMRR)
- Rail to rail voltage swing
- Infinite Power supply rejection ratio

In practice, an OpAmp with zero and infinitive parameters cannot be realized. There are always limitations (e.g., maximum output voltage swing) and trade-offs between the parameters (e.g. the trade-off between open-loop gain and speed) that should be considered during the design process. As a result, there need to be an appropriate specification for each application to device a compromise acceptable for all parameters.

3.2 Real OpAmps

As mentioned in previous section OpAmps cannot be perfect. Due to the circuitry limitations and trade-offs that exist in analogue design there are a number of imperfections in OpAmps which are going to be briefly discussed.

3.2.1 Finite Gain

Open-loop gain is finite in real operational amplifiers. Typically, OpAmps exhibit open-loop DC gain between 50dB to over 60dB (OpAmps with 130db DC gain have been reported [14]). The loop gain of OpAmp placed in a negative feedback loop is large enough, even with typical DC gains, that the circuit gain within the 3-dB bandwidth will be independently determined by the gain of the negative feedback.

3.2.2 Finite Input Impedance

The assumption of infinite input impedance of the OpAmps stems from the assumption of zero input current for the MOSFETs. Typically, the input impedance of the operational amplifier designed with MOSFETs is within the range of 100 to 1000 Mega Ohms.

3.2.3 Non-Zero Output Impedance

Coming to the output impedance, OpAmps can be thought of as voltage sources with internal resistance. The voltage drop across the output impedance of the OpAmp causes power dissipation and delivers less power to the load. The situation is getting worse as the load impedance of the amplifier decreases. However, the use of negative feedback topologies in most applications comes to designer's assistance. Negative feedback lowers the output impedance and reduces output errors accordingly.

Typical output impedance for the open-loop operational amplifiers is in the range of 25-100 Ohms which will be much lower when using the OpAmp in the negative feedback topology (almost a few Ohms). So, the assumption of zero output impedance is quite fair.

3.2.4 Output Swing

Obviously output voltage of the operational amplifier cannot reach to the supply voltages level because of the transistors' overdrive voltages. An amplifier with voltage swing that allows output signal to go very close to supply voltages is called rail-to-rail amplifier.

The need for larger output swing for application like high resolution data converters stems from the necessity of high SNDR and Dynamic Range for such circuits. The output swing limits the linearity of the circuits, especially in low voltage applications. One way to reach high output swing is to use fully differential OpAmps.

3.2.5 Input Current

Input current of the OpAmps includes biasing current and leakage current of the input transistors. Input current for the MOSFETs is much smaller than BJTs or JFETs and it is about a few Pico amperes. Assuming symmetric circuit and matched input current, error will not be introduced to the differential output of the OpAmp but it shows itself as a DC offset and limits the output swing. An OpAmp with a high CMRR can help reducing the offset.

Unfortunately neither the circuit is symmetric nor are the input currents matched. In reality, the input currents could differ by 10 percent or more which produces error to the output

voltage that cannot be tolerated in some applications. The majority of the error could be corrected by the aid of adding a DC path (Ex. through a resistor) to the ground.

3.2.6 Input Offset Voltage

One would expect zero output voltage when applying zero volts to the inputs of a differential amplifier, which is not the case in real amplifiers. The error voltage is caused by some imperfection and mismatches in the internal transistors and resistors of the OpAmp and can be summed up as a DC voltage source and applied in series to one of the inputs of the OpAmp. The input offset voltages can vary from microvolts to milivolts depending on the circuit design and technology. CMOS technology has higher input offset voltage than Bipolar.

In open-loop, due to very high voltage gain of the amplifier, input offset voltage brings the output to its saturation, even though the input voltage is zero. In negative feedback configuration the offset voltage will be amplified along with the input signal, introducing error to the output. This error can be problematic in applications with high precision DC amplification such as high resolution ADCs with very small LSBs (683.6 microvolts for 10-bit ADC and 341.8 micro volts for 12-bit ADC with 1.4 volt peak-to peak differential input voltage).

A major problem regarding to offset voltage is the voltage drift due to temperature change. This problem will be addressed under temperature effects headline.

3.2.7 Common-Mode Gain

In An ideal operational amplifier, common-mode gain is zero and the amplifier amplifies only the differential input signal. However, in real amplifiers the voltages that are common to both inputs are amplified to some extent. This amplification is due to imperfections in tail current sources and mismatches between the transistors and resistors of differential pair. The standard measurement factor created to be used when comparing differential circuits is CMRR which can be calculated using this Equation 3-1:

$$\text{Equation 3-1: } CMRR = \left| \frac{A_{DM}}{A_{CM-DM}} \right|$$

In which the A_{DM} stands for differential gain and A_{CM-DM} stands for common-mode to differential gain. Higher CMRR is desired in OpAmp design.

3.2.8 Power-Supply Rejection

As opposed to the ideal case, supply noises play an important role in real amplifiers. Thus, the performance of amplifier in presence of supply ripples is of concern to many applications, especially mixed-signal applications that often deal with noisy digital supply lines. There is a factor called PSRR which stands for power supply rejection ratio that determines the ability of OpAmps to reject the changes in the power supply.

3.2.9 Noise

Noise exists in amplifiers similar to all analogue circuits. The amount of noise puts a specification for minimum input on the requirement list of the amplifier. If the input signal would be less than this minimum, then it cannot be processed safely. This noise mostly consists of thermal noise and flicker noise of the devices in the circuit. Some of these devices contribute more than others, for example input transistors of the OpAmp. Those devices should be taken care of by widening and applying more bias current.

There is a trade-off between maximum output swing and noise. In order to have more swing, with the same bias current, the overdrive of the transistors can be lowered to allow more swing. As the overdrive voltage goes down, the transconductance of the Amplifier increases which causes more drain noise current. For applications with demands on higher gain or bandwidth, noise becomes an important issue.

3.2.10 Finite Bandwidth

The OpAmp gain calculated at DC will not stay the same at higher frequencies. As the operational frequency of the circuit increases the gain decreases. At first the gain drop is not significant to be considered as system failure, but after the 3-dB frequency the change in DC gain cannot be ignored. In the Figure 3-3 the gain characteristic of an OpAmp is plotted.

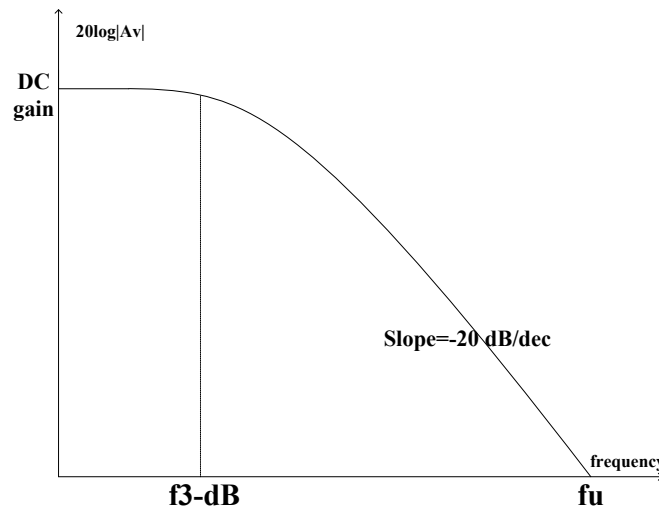


Figure 3-3: Gain versus Frequency

High frequency behaviour of the OpAmp is critical for many applications, especially for those who need high precision gain like the OpAmp hired in a MDAC. The unity-gain frequency of the operational amplifier, “ f_u ” the frequency in which gain drops to zero dB, is a good measure of small-signal bandwidth. Today, using CMOS technology, unity-gain frequencies larger than 1GHz can be achieved.

3.2.11 Nonlinearity

Nonlinearity exists in all analogue circuits including OpAmps. There are several sources that introduce nonlinearity to the circuit. Transistors of the circuit can be considered as one of the main sources as they are inherently nonlinear devices. This source's impact on nonlinearity can be controlled by choosing larger transistor or higher overdrive voltage, especially for input transistors which play a significant role in this case. Considering power and area requirements for circuits, one should be cautious about using these approaches on a large scale. Another source of nonlinearity is the output swing of OpAmps. The output voltage is limited between a minimum and maximum value near the supply voltages. When the output voltage crosses these boundaries, mostly due to high voltage gain, saturation occurs and causes output signal damage. Slewing can be considered as one of the other sources of nonlinearity. Reaching the maximum changing rate, the OpAmp's output voltage will not follow further voltage increase of the input. Internal capacitances are responsible for this effect. The problem can be partly remedied using fully differential circuits in order to suppress second-order harmonics. Furthermore, having higher open-loop gain helps the circuit to have more linearity in closed-loop system.

3.2.12 Stability

The phase difference between input and output leads to oscillation if it becomes 180 degrees in a closed-loop configuration. This means that the amplifier is not stable. Even if the amplifier is stable, it can suffer from ringing which will affect settling time of the OpAmp. To measure stability of an amplifier, the concept of phase-margin comes to assist. PM is defined as the phase difference between 180° and the phase at the frequency in which the loop-gain ($|\beta H(\omega)|$) of the amplifier drops below unity. For PMs above 60 degrees, the step response of the feedback system shows a negligible ringing which provides fast settling time. Higher values for PM, gives more stable systems but not necessarily faster settling time.

For large signal operation, there are other effects that should be considered, such as slewing, output swing and nonlinear behaviour of devices in the circuit in presence of large changes in biasing voltages and currents in the circuit. Therefore, time domain simulations of closed-loop system are more efficient for measuring stability, bandwidth and settling time behaviour of the system.

3.2.13 Temperature Effects

All parameters of MOSFET change with temperature, resulting in changes to the circuit behaviour. Here, a few of those effects are briefly explained.

Temperature has an effect on the threshold voltage of a MOSFET by changing built-in potential [15] $\Phi_F = \Phi_T \ln \left[\frac{N_A}{n_i} \right]$; Where $\Phi_T = \frac{KT}{q}$ is the thermal voltage, N_A is the

substrate doping, n_i is the intrinsic doping parameter for the substrate, q is the charge of an electron and K and T stands for Boltzmann's constant and temperature respectively.

Therefore, the threshold voltage changes according to this Equation 3-2:

$$\text{Equation 3-2: } V_T = V_{T0} + \gamma(\sqrt{|(-2)\Phi_F + V_{SB}|} - \sqrt{|2\Phi_F|})$$

Voltage variation due to temperature change is between $-4 \text{ mV/}^\circ\text{C}$ and $-2 \text{ mV/}^\circ\text{C}$ depending on doping level [16].

Accordingly drain current and voltage of CMOS devices change with temperature, which in turn will affect performance parameters of amplifier, like gain.

One of the most important temperature effects is drifting the input offset. Considering voltage drift as ΔV and temperature change as ΔT , the error is calculated using Equation 3-3:

$$\text{Equation 3-3: } \Delta e = \frac{\Delta V}{\Delta T}; V/^\circ\text{C}$$

It can be seen this error changes linearly with temperature change. Knowing the Δe for an OpAmp, one can determine the voltage drift over a wanted temperature change, which gives a good view of the amplifier to be suitable for target circuit specifications.

3.2.14 Drift

Parameters of semiconductor devices change due to time and temperature changes, resulting in a variation in OpAmp's parameters like: input bias current, offset voltage, etc. these variations are called drift.

3.2.15 Slew Rate

The maximum rate of change of the OpAmp's output is called slew rate. It also means the maximum available current to charge the load capacitor. Slew rate is specified in volts per microsecond ($\text{V}/\mu\text{s}$) and is measured applying a large step to the input and using Equation 3-4:

$$\text{Equation 3-4: } SR = \frac{dV_{out}}{dt} = \frac{I_{max}}{C_L}$$

When applying a step to the amplifier's input, the step response of the feedback system is proportional to the final output voltage of the system. Therefore, when applying larger steps to the input the output change rate will increase, up to the point where the amplifier enters slewing phase. In slewing phase the load capacitor will be charged by the maximum available current in output stage and the change rate will remain constant (SR). Figure 3-4 [11] explains the concept of slewing. It can be seen that increasing the input voltage level wouldn't increase the output change rate after a certain level.

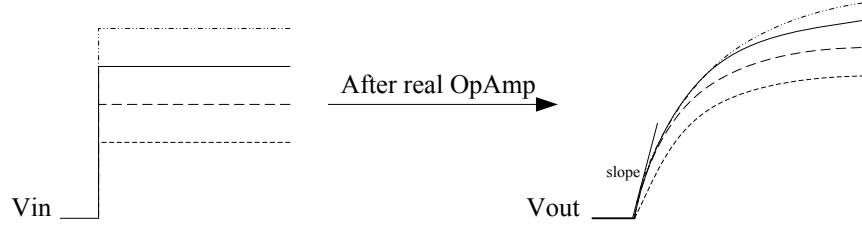


Figure 3-4: Slewing Concept

If output voltage is a sinusoid $V_{out} = V_m \sin(2\pi f_0 t)$, then:

Equation 3-5:
$$\frac{dV_{out}}{dt} = 2\pi V_m f_0 \cos(2\pi f_0 t)$$

So, SR should be greater than the maximum of this derivative:

Equation 3-6:
$$SR > 2\pi V_m f_0$$

3.2.16 Power Considerations

The output current of the OpAmp should be limited to a safe level so as not to damage the OpAmp and following circuitry. The output current also flows through the output impedance, generating heat and increasing temperature. So, if the temperature rises beyond the tolerance of CMOS device, it may cause thermal shutdown or even destroy the OpAmp [17].

3.3 Analogue Design Trade-offs

As discussed, to design an amplifier, there are a lot of issues to be considered. Parameters that can be named are gain, speed, power dissipation and supply voltage, linearity, noise, maximum swing and input and output impedances. These parameters interact with each other and there are trade-offs when optimizing for each parameter. For example, designing to have better noise performance needs the minimum size of transistors to be used which obviously has a conflict of interest as linearity optimization include enlarging devices. Another way to lower nonlinearity is to increase overdrive voltages of MOSFETs which will cause more power dissipation. Figure 3-5 [11] shows the tradeoffs between performance parameters of amplifier.

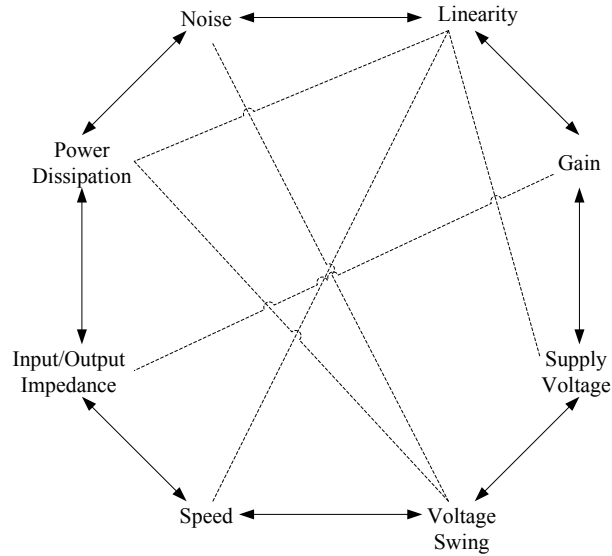


Figure 3-5: Analogue Design Octagon [11]

3.4 OpAmps' Topologies

In this section two topologies of OpAmps are shown. First one is a telescopic topology and the second one is folded-cascode topology. Advantages and disadvantages of each topology are discussed. Then gain boosting and two-stage amplifiers are described. Adding a second stage and gain boosting of cascode devices help to achieve higher gain and also higher voltage swing in second case. At the end a comparison of performance between different topologies of amplifiers will be made.

3.4.1 Telescopic Topology

The first topology to be described here is a 1 stage telescopic amplifier. Telescopic topologies are used to achieve high gain. They increase the gain by boosting output impedance of the amplifier. This structure is also called telescopic cascode configuration. Figure 3-6 shows a fully differential implementation of a cascode OpAmp. To achieve fully differential configuration current-source loads are used which at the same time will help with high gain requirement as well. It is informative to mention that diode-connected loads are used in single-ended output Operational Amplifiers' implementations and they exhibit a mirror pole introduced to the transfer function.

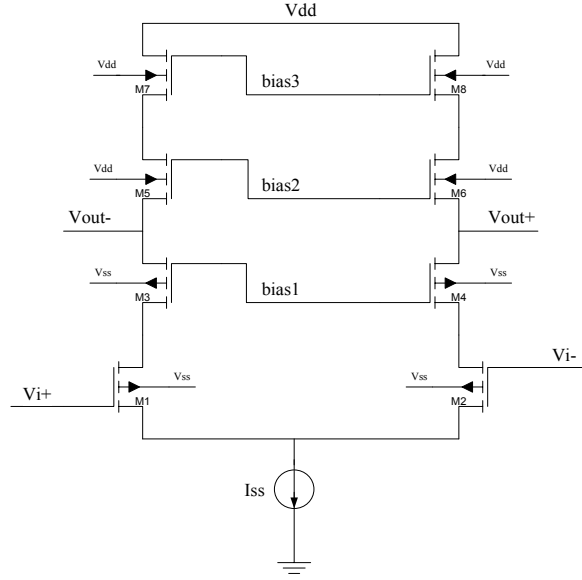


Figure 3-6: Telescopic Amplifier Topology

The output impedance seen from each single output node is equal to:

$$\text{Equation 3-7: } Rout = ([1 + (g_{m3} + g_{mb3})r_{o3}] \times r_{o1} + r_{o3}) \parallel ([1 + (g_{m5} + g_{mb5})r_{o5}] \times r_{o7} + r_{o5})$$

As $G_m \approx g_{m1}$, then the gain can be calculated using Equation 3-8:

$$\text{Equation 3-8: } A_v = G_m \times Rout \approx g_{m1} \times [(g_{m3}r_{o3}r_{o1}) \parallel (g_{m5}r_{o5}r_{o7})]$$

One of the drawbacks of this implementation is the limited output swing. Each transistor cascaded on top of another one, adds an overdrive voltage to the headroom of output branch which will limit the output swing. Therefore, output swing of the fully differential implementation shown in Figure 3-6 is obtained as:

$$\text{Equation 3-9: } Voltage - Swing = 2 \times [V_{dd} - (V_{ov1} + V_{ov3} + V_{ov5} + V_{ov7} + V_{Iss})]$$

Where V_{Iss} is the voltage drop over current source and V_{ov} is overdrive voltage of one transistor in the cascaded branch.

Another drawback is that extra poles are added to the small-signal transfer function of the OpAmp, exacerbating stability issue.

When using this topology, one should be careful about minimum allowable input CM level and choosing bias voltages accordingly. For example, for bias1 we have:

$$\text{Equation 3-10: } CM - level_{\min} = V_{gs1} + V_{Iss} \Rightarrow bias1 = V_{gs3} + V_{ov1} + V_{Iss}$$

3.4.2 Folded-Cascode Topology

We saw that telescopic cascode OpAmps suffer from limited output swing. Folded-cascode OpAmps allow more swing at the output. Although, this topology consumes more power than telescopic topology due to its need for another current source (M3 and M4 act as a current source). This topology can be implemented either employing PMOS input devices or NMOS input devices. Each one has its advantages and disadvantages. In Figure 3-7 and Figure 3-8 two implementation of folded-cascode topology are shown:

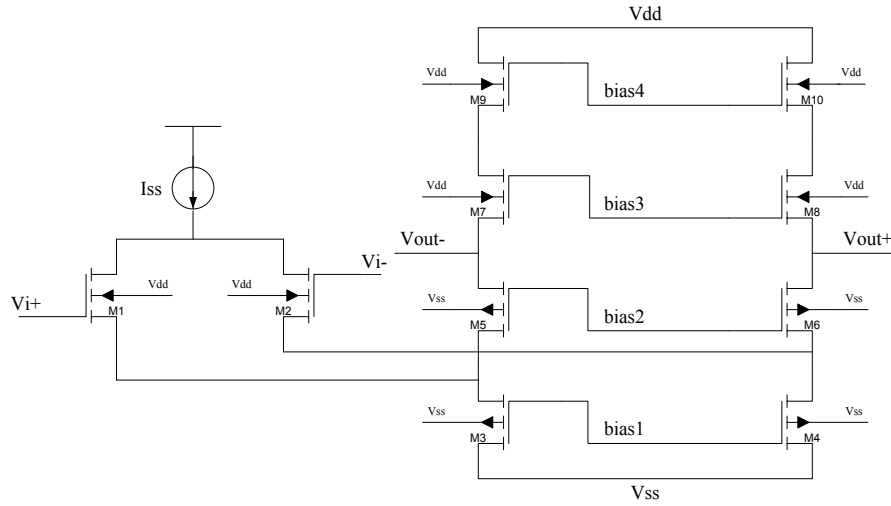


Figure 3-7: Folded-Cascode Implementation Using PMOS Input Devices

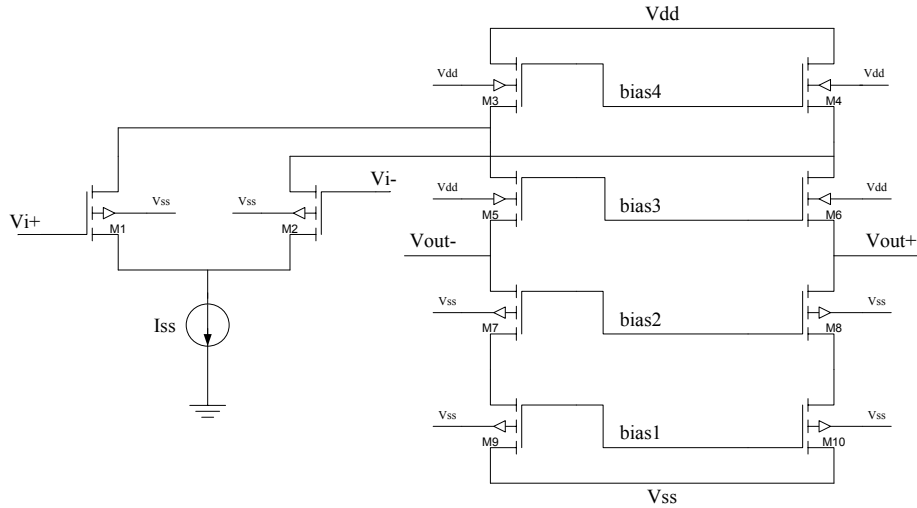


Figure 3-8: Folded-Cascode Implementation Using NMOS Input Devices

It can be seen that voltage swing in folded-cascode topology is higher than telescopic topology by one overdrive voltage across current source ($V_{I_{ss}}$). So in the circuit of Figure 3-7:

Equation 3-11: $Voltage - Swing = 2 \times [V_{dd} - (V_{ov3} + V_{ov5} + V_{ov7} + V_{ov9})]$

Using the same approach as for telescopic OpAmp, the gain of folded-cascode topology can be obtained as:

Equation 3-12:

$$R_{out} = ([1 + (g_{m5} + g_{mb5})r_{o5}] \times (r_{o3} \parallel r_{o1}) + r_{o5}) \parallel ([1 + (g_{m7} + g_{mb7})r_{o7}] \times r_{o9} + r_{o7})$$

Equation 3-13: $A_v = G_m \times R_{out} \approx g_{m1} \times [(g_{m5}r_{o5}(r_{o1} \parallel r_{o3}) \parallel (g_{m7}r_{o7}r_{o9}))]$

This gain is about 2-3 times less the gain of telescopic OpAmps. One reason is the lower transconductance of PMOS input devices compared to NMOS input devices. Another reason is appearing of r_{o1} in parallel with r_{o3} , which will reduce the output impedance of amplifier.

Another issue that should be kept in mind is that the pole in the source of cascode devices (M5 and M6 in Figure 3-7) is closer to the origin than that of telescopic OpAmp (M3 and M4 in Figure 3-6). In folded-cascode structure, capacitance in the mentioned node is made of C_{gs5} , C_{sb5} , C_{gd3} , C_{db3} , C_{gd1} and C_{db1} , which has two more elements (C_{gd3} and C_{db3} , other elements are from input and cascode devices which exist in both structures) than that of telescopic structure. This issue is exacerbated when using NMOS input devices. The reason lies within the need for larger PMOS transistors, as second current source, to carry both currents of input and cascode devices and obviously larger devices contribute more capacitance. Also lower transconductance of PMOS transistors, as cascode devices, increases the impedance of the node ($R_{node} \approx \frac{1}{g_{m5} + g_{mb5}}$ in Figure 3-8), which will help

to bring pole to lower frequencies.

One of the important benefits of folded-cascode OpAmps is that their input CM level range is larger than that of telescopic OpAmps. Depending on the kind of input device, input CM level can be very close to one of the supply sources. In case of PMOS input devices, input CM level can be zero and having NMOS input device, OpAmp tolerate input CM level equal to V_{dd} . In general the choice of input device depends on the application. Whether gain is the target or CM level dictates the input device.

3.4.3 Gain-Boosting

In telescopic and folded-cascode topologies, increasing output impedance has been used as a means of increasing gain. In both, stacking more transistors in output branch as cascode devices helps to do so. What if there is a need for higher gain and larger output swing at the same time? Then, there would be no good outcome, inserting another level of transistors in the stack. The idea behind gain boosting is to increase the output impedance further more to achieve higher gain without adding more transistors to the output branch and reducing the swing as a result. In this approach, the cascode device is placed in a current- voltage feedback [11] using an amplifier. Assuming telescopic OpAmp, both NMOS cascode devices in signal path and the PMOS cascode devices in the load current

source can be used for gain boosting. In Figure 3-9 cascode transistors M3-M6 are placed in the feedback loop:

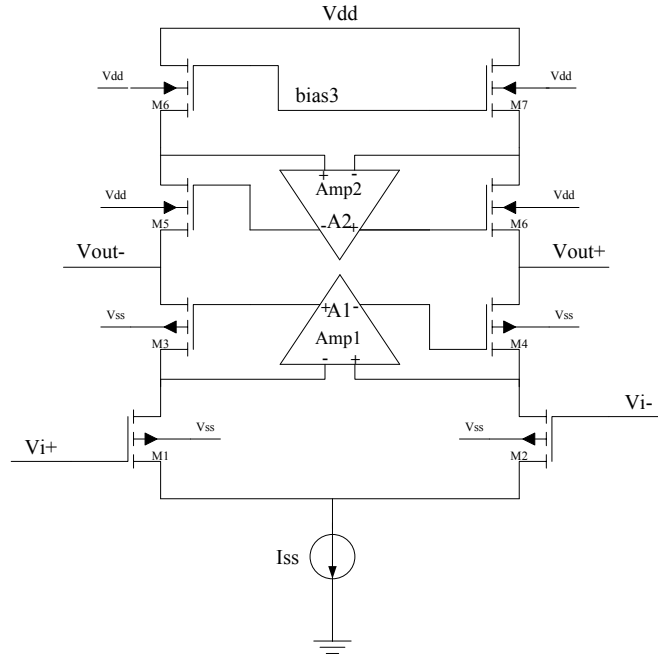


Figure 3-9: Gain Boosting Applied to Telescopic OpAmp Topology

The output impedance, using gain boosting technique, is obtained from parallel calculation of impedances seen by looking into drain of cascode devices, as before, but this time multiplied by the gain of auxiliary amplifiers A_1 and A_2 :

Equation 3-14:

$$R_{out} = ([1 + (g_{m3} + g_{mb3})r_{o3}A_1] \times r_{o1} + r_{o3}) \parallel ([1 + (g_{m5} + g_{mb5})r_{o5}A_2] \times r_{o7} + r_{o5})$$

As $G_m \approx g_{m1}$, then the gain can be calculated:

$$\text{Equation 3-15: } A_v = G_m \times R_{out} \approx g_{m1} \times [(A_1 g_{m3} r_{o3} r_{o1}) \parallel (A_2 g_{m5} r_{o5} r_{o7})]$$

So, the gain is enhanced by increasing output impedance. The auxiliary amplifier can have any topology from a simple CS amplifier to fully differential folded-cascode topology. The sensing transistor at the input of these amplifiers should be chosen, such that they suit the CM level of the voltage being sensed (source of cascode devices). For example in Figure 3-9, if the Amp1 is a fully differential folded-cascode, the input devices should be PMOS as they can tolerate almost zero input CM level and the source voltage of $M_3 + M_4$ will become as low as $V_{ov1} + V_{I_{SS}}$. Similarly, NMOS input devices are more suitable to be used in Amp2 as they will sense input CM level close to V_{dd} .

An important issue to remember is that, although the OpAmp is still a 1-stage amplifier, the poles in auxiliary amplifier will affect the transfer function. The effect wouldn't be very dramatic as the path is not the feed-forward path where most of the signal will flow through it. Nevertheless, one should be careful with stability around the extra loops created by gain boosting as well as stability around the main loop, placing the amplifier in closed loop configuration, when designing the circuit.

3.4.4 Two-Stage OpAmps

Two-stage OpAmps are used for their ability to provide more gain and swing. Basically, the second stage provides about 5-15 dB gain, which is not very high. But the higher output swing provided by the second stage is crucial to some applications, especially with lower supply voltages in today's technologies. So, the second stage is a simple amplifier like a CS stage, as shown in Figure 3-10 below:

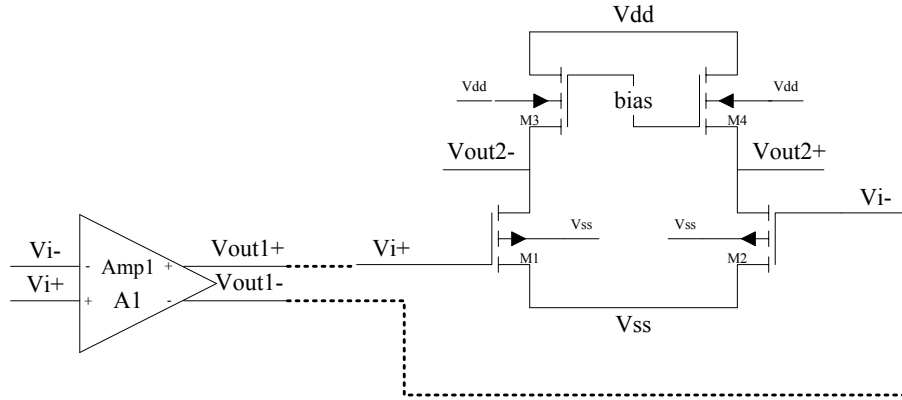


Figure 3-10: Two- Stage OpAmp

The second stage's gain is multiplied by the gain of the first stage:

$$\text{Equation 3-16: } A_{v-total} = A_{v1} \times A_{v2} = A_{v1} \times [g_{m1}(r_{o1} \parallel r_{o3})]$$

The second stage's swing is much larger than say a telescopic output swing:

$$\text{Equation 3-17: } Voltage - Swing = 2 \times [V_{dd} - (V_{ov1} + V_{ov3})]$$

The output stage's current should be high for the sake of speed, but, not that high to damage MOSFET devices or produce too much thermal noise. Obviously power dissipation should be kept under control too.

3.4.5 Comparison between Different Topologies of OpAmps

In this section, we sum up properties of different amplifier topologies that have been discussed in previous sections.

Telescopic OpAmps have high speed as the input device's current flows directly into output impedance, but they suffer from limited output swing. This topology is simple and there is only one current source in it, so they dissipate power less than other topologies.

Folded-cascode OpAmps stand next in the line. Compared to the telescopic topology they have less gain and speed and dissipate more power. But they have found their place in a wide range of applications due to their larger output swing and their extended input CM level range.

Gain boosting and adding second stage is two powerful design schemes to obtain higher gain and in second case higher swing as well. As the level of the complexity of the circuit goes up, power consumption increases.

Bellow in Table 3-1 [11] all properties of different topologies have been compared with each other:

Table 3-1: Comparison between Performance of Different OpAmp Topologies [11]

	Gain	Output Swing	Speed	Power Consumption	Noise
Telescopic	Medium	Medium	Highest	Low	Low
Folded-Cascode	Medium	Medium	High	Medium	Medium
Gain Boosted	High	Medium	Medium	High	Medium
Two-Stage	High	Highest	Low	Medium	Low

In this work, a two-stage gain boosted amplifier is designed to achieve high DC-gain and output swing. The price to be paid is high power consumption which is not avoidable when a high performance amplifier is needed.

Chapter4. Designed OpAmp

In this chapter requirements of an OpAmp to be employed in a 12-bit pipelined ADC are discussed and calculated. After that the designed OpAmp and its performance metrics are shown. ADC structure and the use of designed OpAmp in the ADC will be described in next chapter.

4.1 OpAmp Requirements

For an OpAmp-based design of a high resolution and high speed pipelined ADC, there are high requirements for the OpAmp design to be satisfied. These two definition “high resolution” and “high speed” for an ADC adds a great deal of challenge on the OpAmp design to achieve the required performance regarding DC-gain, Bandwidth, noise, stability, speed and swing. All of which should be achieved under critical conditions of decreased supply voltages and intrinsic gain of today’s CMOS technology. The down sized transistors of new coming technologies also have higher leakage and lower output resistance. They are faster switches as a result of the reduced parasitic capacitances (due to reduced transistor dimensions). Because of the higher number of transistors in smaller area, heat production is another problem of scaling in new technologies which will cause slower operation and reduced reliability and lifetime of the transistors. These transistors are also more prone to process variation. All of these characteristics of new scaled down technologies add more error to the OpAmp’s transfer function, making it harder to satisfy the stringent requirements on the OpAmp.

OpAmps are the basic building block of an ADC which determine the speed and accuracy of the ADC. They introduce gain error and nonlinearity which should be minimized in design process or compensated for by digital correction circuitry. They are also the most power hungry part of the ADC and dissipate almost 60-80% of the total power. There are a few techniques to reduce OpAmps power consumption [18], like using class AB amplifiers which only consumes dynamic current, OpAmp sharing and OpAmp current reuse.

As discussed inChapter2, the OpAmp is used in the 2.5 b MDAC structure of the pipelined ADC. The OpAmp is placed in a negative feedback with amplification factor of 4. Now it is time to see what the requirement specification for this OpAmp is. Here we discuss DC-gain, Gain-Bandwidth (GBW), Slew-Rate (SR) and Noise.

Before extracting the requirements for an OpAmp, there are a few parameters from the specification of the ADC that are needed in the calculations:

Equation 4-1: $V_{ref-ADC} = 700mv \Rightarrow V_{in-FS}(Diff) = 1.4v$

Equation 4-2: $V_{in-FS}(Diff) = 1.4v \Rightarrow LSB = \frac{1.4v}{2^{12}} = 342\mu v$

Equation 4-3: $\frac{1}{2} LSB = 171\mu v$

4.1.1 DC-Gain

OpAmps finite gain introduces error to the next stage in the pipelined ADC. For each stage, this error should be less than LSB voltage of an ADC with the same reference voltage and resolution of remaining bits (total bits that the whole ADC resolves minus resolved bits). To achieve the accuracy requirement for an N-bit pipelined ADC, the gain error contributed by the first stage with M-bit resolution should satisfy the equation below:

$$\text{Equation 4-4: } \left(\frac{1}{\beta} - \frac{A}{1 + A\beta} \right) \frac{V_{ref}}{2^M} \leq \frac{V_{ref}}{2^{N-M}}$$

Where $\frac{1}{\beta}$ is the inter-stage gain factor and equals to 2^{M-1} for 2.5 bps structure. Equation 4-4 can be simplified to:

$$\text{Equation 4-5: } A \geq 2^{N-1}$$

For N=12, M=3 and $\frac{1}{\beta}=4$, the minimum open-loop gain of 66.22dB is needed. To have some margin open-loop larger than 66dB is considered to take care of process variation and mismatch errors. Some of the gain and offset errors of the OpAmp is corrected employing digital correction logic.

4.1.2 Gain-Bandwidth (GBW)

The gain-bandwidth product of an amplifier is the product between its bandwidth and the gain at which the bandwidth is measured. This means that at unity frequency (f_u), at which the gain of amplifier falls to unity ($A_v|_{f_u} = 0dB = 1$), the GBW product of the amplifier equals unity frequency ($GBW = gain \times bandwidth = 1 \times f_u = f_u$). This also means that if the OpAmp is placed in a unity-gain feedback, the bandwidth at which the gain remains one equals f_u . With the same GBW product, the same amplifier if placed in a negative feedback system to achieve a gain of 4 will have a bandwidth equal to (Equation 4-6):

$$\text{Equation 4-6: } GBW = gain \times bandwidth = 4 \times bandwidth \Rightarrow bandwidth = \frac{GBW}{4}$$

In the same manner the DC-gain of the OpAmp can be calculated at 1Hz bandwidth:

$$\text{Equation 4-7: } GBW = gain \times bandwidth = DC - gain \times 1Hz \Rightarrow DC - gain = \frac{GBW}{1Hz}$$

Having said all that, it is important to remember that these calculations will not give the exact values for circuits with two poles or more. That is because the GBW product is

independent of gain, in which it is measured, only in one pole circuits. Nevertheless, GBW product provides useful insights even into complex circuits. At the end all performance metrics of the circuit should be verified by simulation.

To achieve high accuracy requirements for N-bit pipeline ADC, unity frequency should be much larger than sampling frequency. Using same method for calculating minimum DC-gain, the error produced by low **bandwidth of the OpAmp should be less than 1 LSB** of an ADC with the same reference voltage and resolution of remaining bits (total bits that the whole ADC resolves minus resolved bits):

$$\text{Equation 4-8: } \left[\frac{1}{\beta} - \frac{1}{\beta} \times \left(1 - e^{-\frac{t}{\tau}} \right) \right] \frac{V_{ref}}{2^M} \leq \frac{V_{ref}}{2^{N-M}}$$

In Equation 4-8, $\frac{1}{\beta}$ stands for inter-stage gain, V_{ref} is the reference voltage of ADC,

$1 - e^{-\frac{t}{\tau}}$ is the step response of feedback system with $\tau = \frac{1}{2\pi \cdot \beta \cdot f_u}$. Equation 4-8 can be simplified to:

$$\text{Equation 4-9: } f_u > \frac{(N-M)\ln(2)}{2\pi \cdot \beta \cdot T_{settle}}$$

Considering 300MHz sampling frequency ($T_{sampling} = 3.33ns$) and according to clocking scheme of the designed ADC, the time available for settling is:

$$\text{Equation 4-10: } T_{settle} = \frac{3}{4} \times T_{sampling} = 2.5ns$$

For N=12, M=3 and $\beta=1/4$, the minimum gain-bandwidth required is 1.59 GHz.

4.1.3 Slew-Rate (SR)

Slew-rate is the **maximum rate of output voltage changes** and it is calculated using the **maximum available current to charge the load capacitance**. It is one of the sources of distortion when the output is near its maximum swing.

$$\text{Equation 4-11: } SR > \frac{dV_{max}}{dt} = \frac{Voltage_{swing-Diff} \times 2\pi}{T_{settle}}$$

For a voltage swing of 1.4v (full scale differential voltage range of ADC) and sampling rate of 300 MHz (considering half of the sampling period for settling), SR should be greater than $5.278 \frac{kv}{\mu s}$.

4.1.4 Noise

According to the noise budgeting in Chapter2 the maximum allowable noise power at the output of the first stage of pipelined ADC is $435 \mu\text{V}$. 50% of this voltage is reserved for thermal noise and this will leave $217 \mu\text{V}$ for other noises including jitter noise, OpAmp noise, etc. large portion of this noise is due to amplifiers noise, so, $150 \mu\text{V}$ of this power is reserved for it. This amount of noise power equals -76.5dB , demanding for an SNR value higher than 70.5dB .

4.1.5 Summary of OpAmp's Requirements

In the Table 4-1, the needed requirements on the OpAmp to be used in the pipelined ADC are summarised.

Table 4-1: Summary of OpAmp's Requirements

Performance Metrics	Required Values
f_u	2.12 GHz
Slew Rate	$5.278 \text{ kv}/\mu\text{s}$
DC – gain	67 dB
SNR	71dB

4.2 Designed OpAmp

The designed OpAmp is a two-stage, fully differential, Cascode current mirror topology modified for low-voltage operation. It is an extended version of OpAmp used [19]. In Figure 4-1 the architecture of the amplifier is shown:

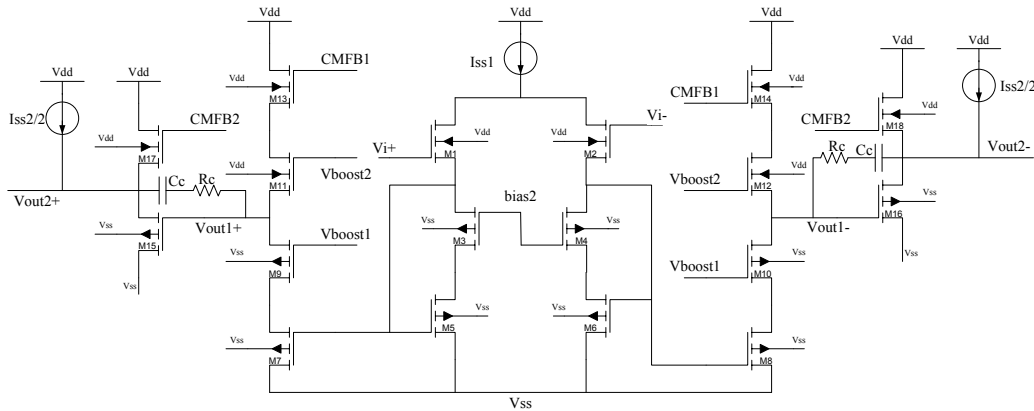


Figure 4-1: OpAmp Architecture

The OpAmp is a **two-stage amplifier** to achieve high gain and voltage swing. It is also uses gain boosted cascode devices. Input devices are chosen to be PMOSFETs because of their lower flicker noise and more flexibility about the input CM level. Second stage incorporates NMOS devices for their higher intrinsic gain. Second stage is a simple CS

stage to allow more output swing. 2pF load capacitor is considered to simulate next stage's input capacitance. The compensation scheme used here is Miller Compensation.

4.2.1 Common-Mode Feedback (CMFB)

Both stages CM levels are regulated by common-mode feedbacks. Figure below represents CMFB circuit:

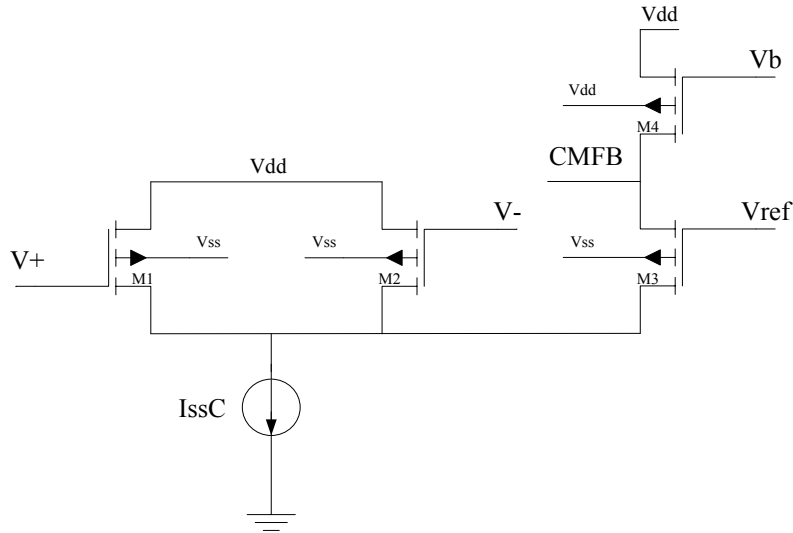


Figure 4-2: CMFB Circuit

Differential outputs of each stage are sensed by a differential pair and compared to a voltage reference. In case of any differentiation, the CMFB brings back the output CM mode level to its equilibrium.

4.2.2 Boosting Amplifiers

Boosting amplifiers are folded-cascode OpAmps. As explained in Chapter3, folded-cascode OpAmps have high voltage swing and moderate gains. They also allow more input CM mode range. The 4 stacked transistors of first stage and the boosting amplifiers placed in the main amplifiers circuit are shown in Figure 4-3.

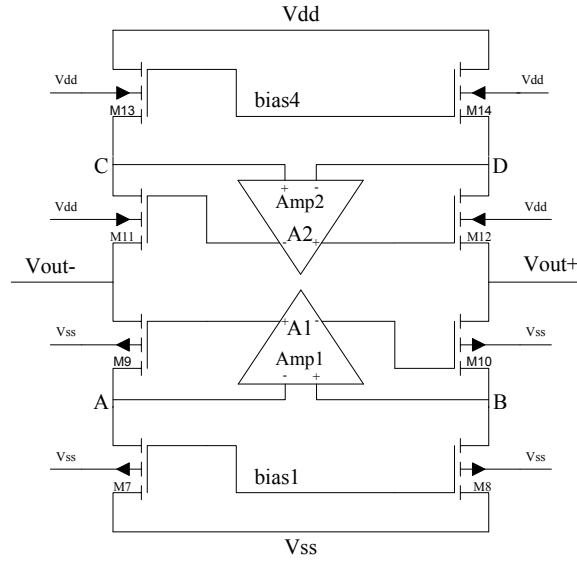


Figure 4-3: Boosting Amplifiers Placed in The First Stage's Output Branch

Amp1 senses voltages of points A and B, regulates the cascode devices' (M9-M10) gate-source voltages and amplifies the total gain by A1. Amp1 has PMOS input devices to deal with low voltage CM levels in A and B. Amp2 incorporates NMOS input devices due to same reasoning. The architecture of boosting amplifier number two is shown below:

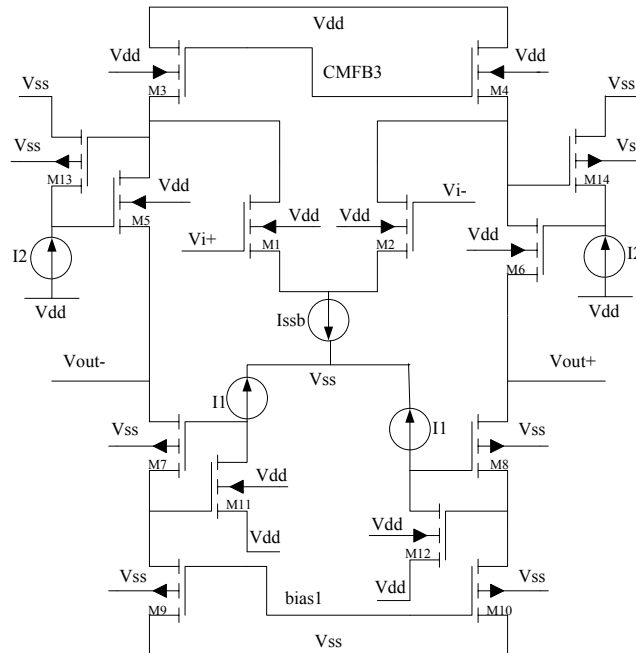


Figure 4-4: Boosting Amplifier

Boosting amplifiers are gain boosted as well. The technique is called Nested-boosting [14]. Sometimes the second boosting amplifier is simple, like this case, but it also can be scaled

version of main boosting amplifier if more gain is needed. One should be cautious when putting boosting amplifiers into the circuit as they introduce internal loops that can be unstable. To check for stability around internal loops, probes (to break the loop during simulation) and stability simulation can be used. In Figure 4-5, Figure 4-6, Figure 4-7 and Figure 4-8 gain and phase plots of both boosting amplifiers are shown.

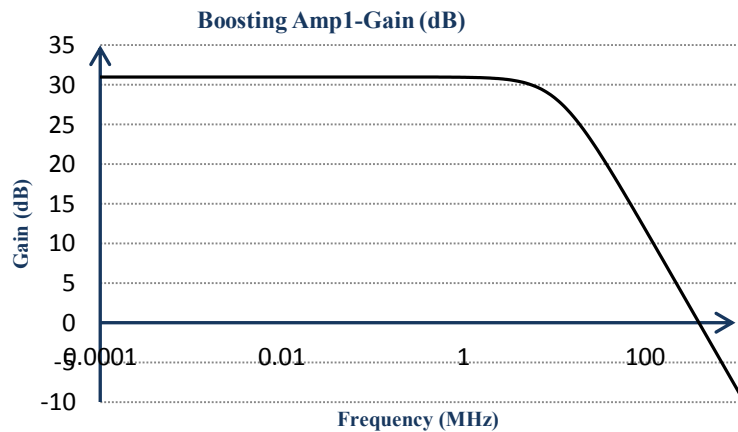


Figure 4-5: Boosting Amp1 Gain Plot

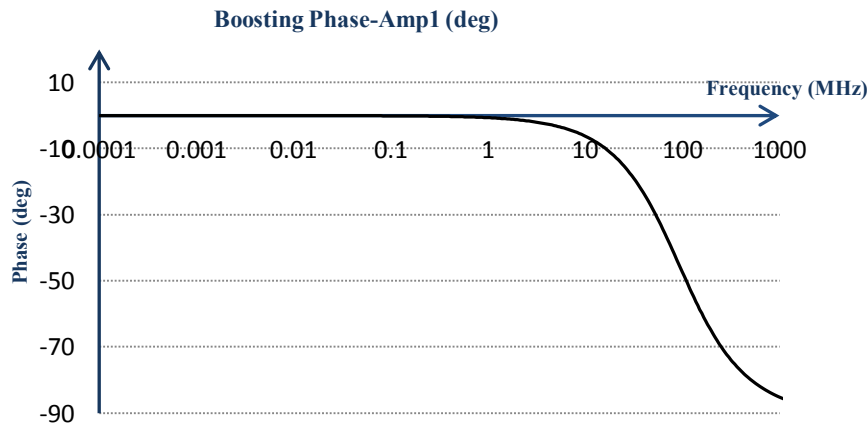


Figure 4-6: Boosting Amp1 Phase Plot

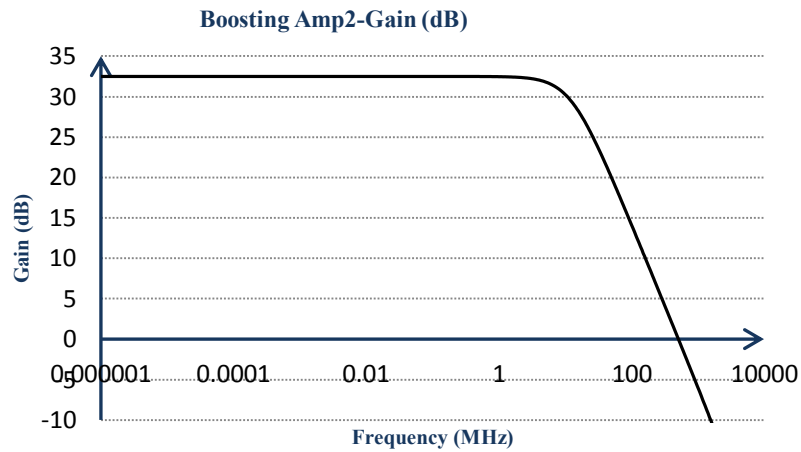


Figure 4-7: Boosting Amp2 Gain Plot

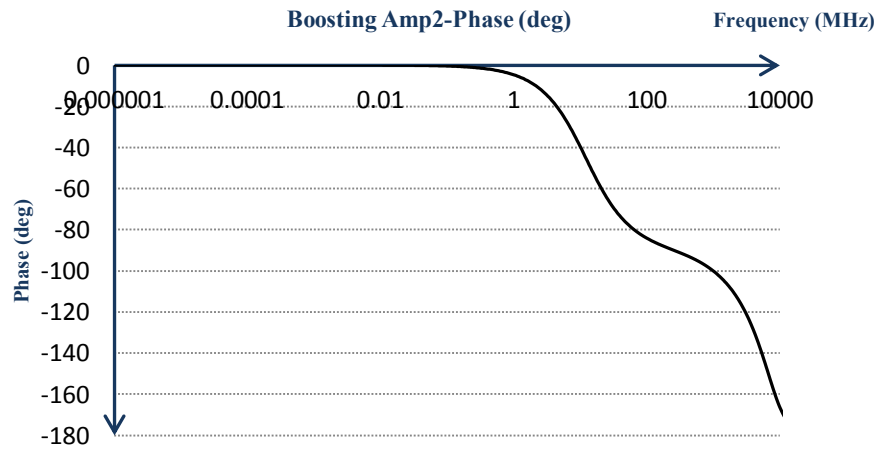


Figure 4-8: Boosting Amp1 Phase Plot

Table 4-2 and Table 4-3 contains simulated parameters for boosting amplifiers:

Table 4-2: Boosting Amplifier No.1 Results

Performance Metrics	Simulated Value
f_u	393.4 MHz
BW_{3dB}	11.45 MHz
$DC - gain$	30.98 dB
PM	89.93 deg
$i_{V_{dd}}$	11.44 mA

Table 4-3: Boosting Amplifier No.2 Results

Performance Metrics	Simulated Value
f_u	512.9 MHz
BW_{3dB}	12 MHz
$DC - gain$	32.48 dB
PM	94.25 deg
$i_{V_{dd}}$	15.5 mA

4.3 Test Bench

The OpAmp is placed in the test bench illustrated in Figure 4-9 to be simulated and its parameters are calculated. Sampling capacitor is 4 times of feedback capacitor to provide gain of 4 for the MDAC ($C_s/C_f = 4$). The noise voltage at the output of amplifier falls below 0.2LSB by Choosing C_s larger than 100fF for 10-bit ADC and 2pF for 12-bit ADC. C_l is chosen 2pF to simulate the load effect of next stage.

The probe module shown in the figure below is placed in the feedback to break the loop when using the stability simulator (stb Analysis in Analog Design Environment) in Cadence. The stability simulator calculates loop-gain and loop-phase and is used to determine stability of the circuit around the loop.

The analogue writer module is responsible for sampling the output data and dumping the sampled data into a text file. The text file can be read by Matlab and used for OpAmp's performance determination. The writer is written in VerilogA Code and the Code can be found in Appendix B.

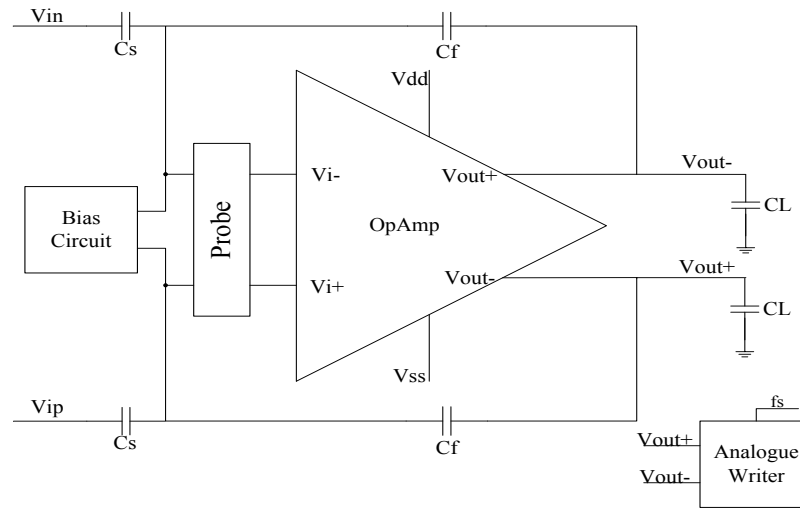


Figure 4-9: OpAmp Test Bench

4.4 Designed OpAmp's Result

The OpAmp reaches 72dB DC-gain and the gain stays constant when the output swing increases up to the point that the output voltage clips. In Figure 4-10 and Figure 4-11 the open-loop gain and phase of the designed OpAmp is shown:

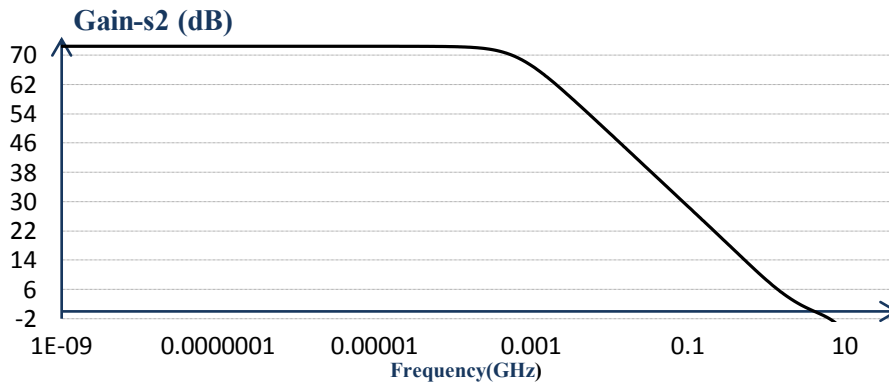


Figure 4-10: Open-Loop Gain Plot of 2-stage, Gain Boosted OpAmp

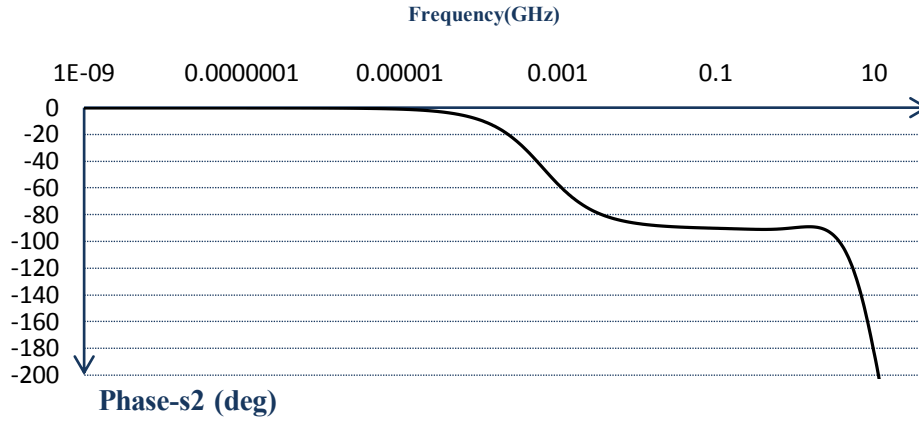


Figure 4-11: Open-Loop Phase Plot of 2-stage, Gain Boosted OpAmp

Other performance metrics of the amplifier are simulated and summarized in Table 4-4:

Table 4-4: OpAmp Simulated Performance Metrics

<i>Performance Metrics</i>	<i>Simulated Value</i>
f_u	4.077 GHz
BW_{3dB}	640.4 KHz
$DC - gain$	72.35 dB
PM	76.01 deg
$i_{V_{dd}}$	123.3 mA
$SNR(\text{for } fs \text{ up to } 320 \text{ MHz})$	>100dB
$Slew Rate$	22.5kv/ μs

Settling behaviour of the OpAmp, for being used in 12-bit or 10-bit pipelined ADC and placed in the 2.5 bps MDAC (amplification factor of 4) is verified. The simulation is done applying low frequency (1MHz) pulse waves with different amplitudes to the input and recording the settling time of the amplifier, when the OpAmp settles to half of the corresponding LSB. For example in Figure 4-12 an input signal's rising edge with peak to peak voltage of 200mv along with the output signal's rising edge is shown.

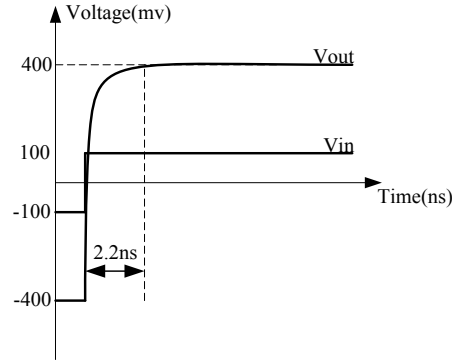


Figure 4-12: OpAmp's Input/output Pulses' Rising Edge

LSB for 12-bit ADC with maximum 1.4v input voltage (differential peak to peak voltage) is $342\mu\text{v}$ and settling between the $171\mu\text{v}$ (half LSB) error from final value is evaluated.

Table 4-5: Settling Time of The OpAmp for Being Placed in 12-bit ADC

$V_{in-pp-Diff}(mv)$	$V_{out-pp-Diff}(mv)$	$T_{settling}(ns)$	$f_{s-max}(MHz)$
200	800	2.243	356.66
240	960	2.92	273.97
280	1120	3.27	244.65
320	1280	3.56	224.71
360	1440	3.86	207.25

LSB for 10-bit ADC with maximum 1.4v input voltage (differential peak to peak voltage) is 1.37mv and settling between the $683.6\mu\text{v}$ (half LSB) error from final value is evaluated.

Table 4-6: Settling Time of The OpAmp for Being Placed in 10-bit ADC

$V_{in-pp-Diff}(mv)$	$V_{out-pp-Diff}(mv)$	$T_{settling}(ns)$	$f_{s-max}(MHz)$
200	800	2	400
240	960	2.25	355.55
280	1120	2.57	311.28
320	1280	2.79	286.74
360	1440	2.84	281.7

In order to calculate signal and noise power and calculate SNR, an analogue writer is used to sample the OpAmp's output sine and dump the sampled data into a text file. The text file can be read by a Matlab code and used to calculate SNR. If frequency of the output sine wave is considered as f_i and analogue sampler's clock frequency as f_s then they should relate to each other according to Equation 4-12 to satisfy coherent sampling:

Equation 4-12:
$$f_s = f_i \times \frac{M_{sample}}{N_{cycle}}$$

In the equation above, M_{sample} stands for number of samples or size of the FFT (here it is equal to $4096=2^{12}$) and N_{cycle} stands for odd or prime number of the complete cycles of output sine. SNR Value is calculated for different input frequencies and amplitudes and its value stays above 100 dB for frequencies up to 320MHz.

The designed OpAmp is placed in an ideal model of the pipelined ADC, where every other block in the ADC is written in VerilogA. In Chapter5, the simulation result for an ideal pipelined ADC incorporating the schematic OpAmp is shown.

4.5 Comparison with other works

In Table 4-7 the OpAmp designed in this thesis is compared with other works' designed OpAmps. All these OpAmps are designed to be used in pipelined ADCs. In the table the resolution and sampling frequency of the pipelined ADC, for which the OpAmps are designed, are shown. As can be seen, OpAmps are designed in different CMOS technologies which will affect the achievable results as discussed earlier in this chapter. The strongest aspect of this thesis is the high GBW (4.077 GHz) which is achieved in 65nm CMOS technology.

Table 4-7: Comparison between the OpAmp's results and other works

	This work	[19]	[10]	[18]	[22]
Technology	65nm CMOS	rad –tolerant CMOS	90nm CMOS	0.18um1P6M CMOS	0.13um CMOS
ADC Resolution	12	12	12	12	8
ADC Sample-Rate	300MHz	40MHz	200MHz	50MHz	440MHz
DC Gain/ Feedback-Gain	72.35dB/4	90dB/4	28dB/1.71	74.2dB/4	60dB/2
GBW	4.077GHz	260MHz	-	-	3GHz
Power	135mW	3.9mW	192mW	-	-
PM	76.01deg	-	-	-	60deg

Chapter5. Simulation Result of Pipelined ADC Incorporating Designed OpAmp

In this chapter, three models of pipelined ADC are introduced (i.e. high level model, high level model with the designed OpAmp inserted in the circuit and finally the pipelined ADC in schematic), simulated and the result is shown. First of all, a completely high level model based on the pipelined ADC architecture which is designed in schematic is introduced and simulated to obtain the performance metrics of an ideal pipelined ADC. These metrics are used as reference values to be compared with the performance metrics of the later versions of the ADC which incorporate blocks designed in schematic replacing their high level counterparts.

5.1 Simulation Result for the High Level Pipelined ADC

High level model of the pipelined ADC is developed using VerilogA. SNR, SFDR, SNDR, THD and ENOB are calculated by Matlab and the data used in Matlab is dumped in a text file by a writer in cadence. Coherent sampling is applied and 4096 (2^{12}) samples are collected during simulation for different sampling frequencies and input amplitudes of a sine wave. The simulation results are depicted below.

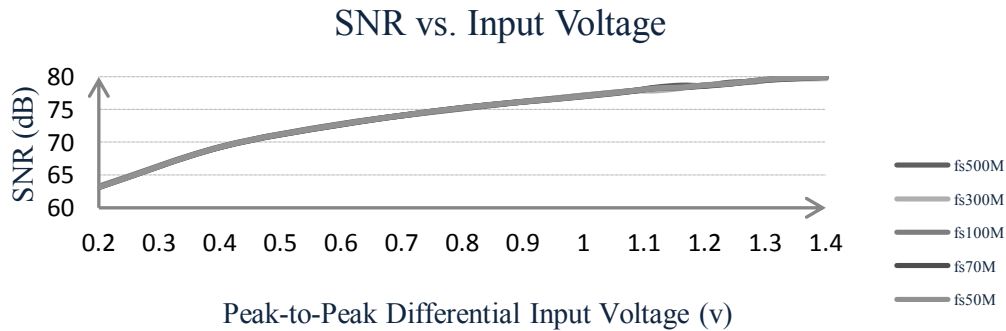


Figure 5-1: SNR vs. Peak-to Peak Differential Input Voltage Plot for Ideal ADC

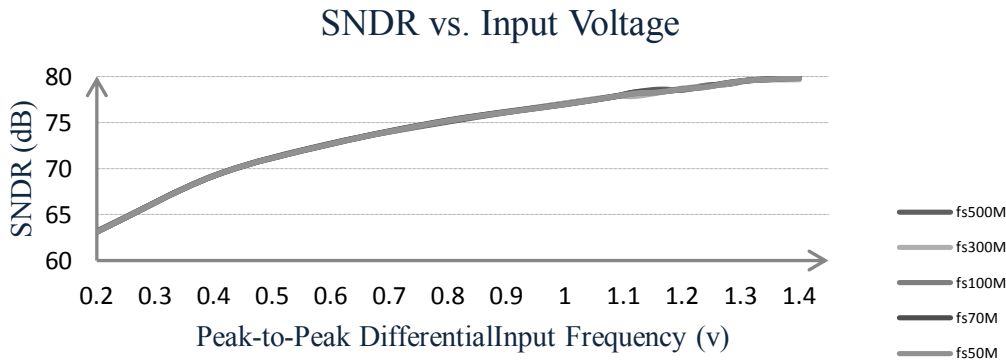


Figure 5-2: SNDR vs. Peak-to Peak Differential Input Voltage Plot for Ideal ADC

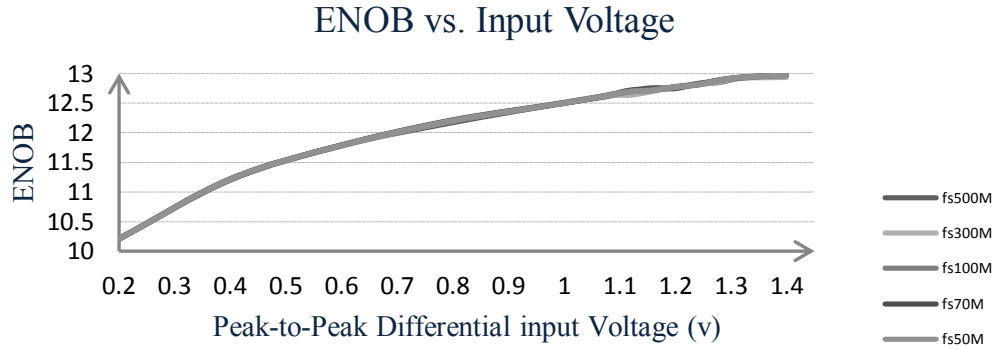


Figure 5-3: ENOB vs. Peak-to Peak Differential Input Voltage Plot for Ideal ADC

The high level model's simulation results are illustrated in figures above (from Figure 5-1 to Figure 5-3). Two sets of simulations are performed to study the effect of input voltage amplitude and sampling frequency changes on ADC's performance. First set of simulations are based on changes in input voltage amplitude where the sampling frequency is kept constant and another set are based on changes in sampling frequency where the input voltage amplitude is kept the same. Both sets of simulations are repeated for different constant values. For example in Figure 5-1 the effect of input voltage amplitude changes (from 800mv to 1400mv) on SNR is studied where the sampling frequency is 100MHz and this simulation set is repeated for other sampling frequencies. The results indicate that for high level model of the pipelined ADC, the voltage amplitude of the input plays the most important role in determining the ADC's performance. As the input signal's Voltage amplitude (signal power) increases, better performance values are obtained. In this Model, sampling frequency does not affect ADC's performance that much and it can be seen that all the curves related to different sampling frequencies in the SNR, SNDR and ENOB plots for Ideal ADC are placed on top of each other. Obviously, the reason is that the ideal blocks in the pipelined ADC's architecture do not need a lot of time to settle or prepare the result after the clock's rising edge. It is worthwhile to say that the sampling frequency changes will affect the performance at higher frequencies. That would be because of the delays, rising time and falling time which are assigned to the clocks and signals of the ideal blocks in the ADC's architecture in order to make the ideal model's behaviour more similar to real one.

5.2 Simulation Result for the High Level Pipelined ADC with the OpAmp in Schematic

This model of the pipelined ADC is similar to the high level model except for the inter-stage gain block which is replaced by the designed OpAmp in schematic. The OpAmp is placed in a closed-loop configuration with a feed-forward gain of 4. Similar to the ideal model, SNR, SFDR, SNDR, THD and ENOB are calculated and the simulation results are shown below.

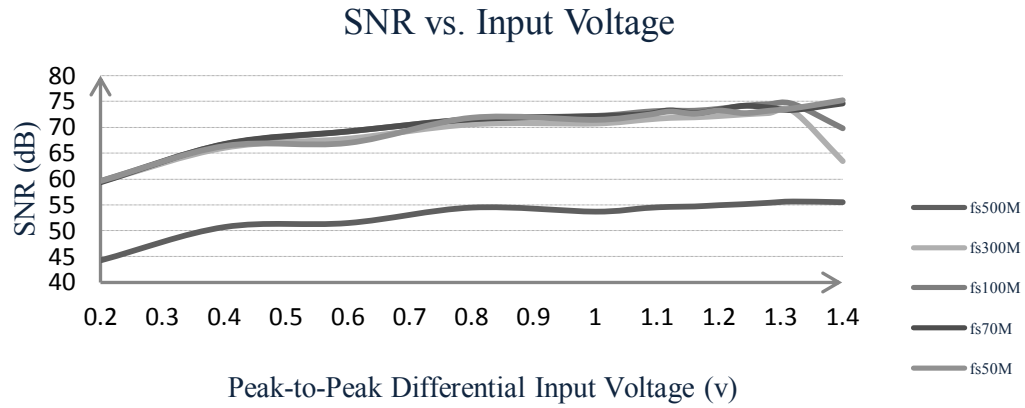


Figure 5-4: SNR vs. Peak-to-Peak Differential Input Voltage Plot for Ideal ADC with Transistor Level OpAmp

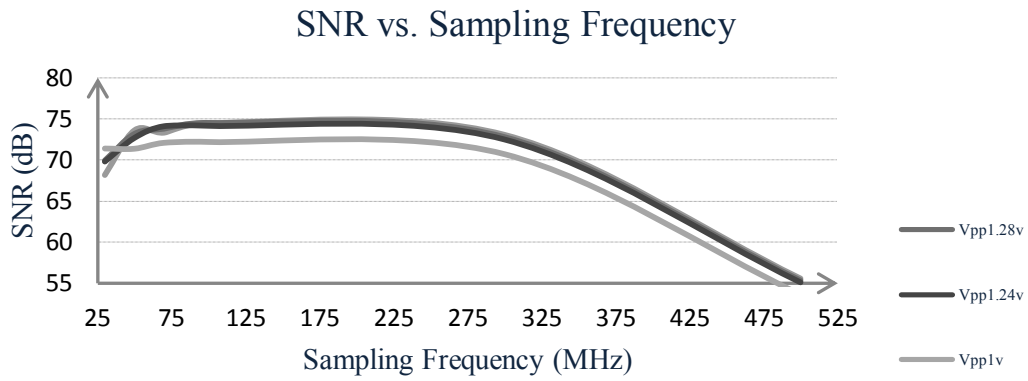


Figure 5-5: SNR vs. Sampling Frequency Plot for Ideal ADC with Transistor Level OpAmp

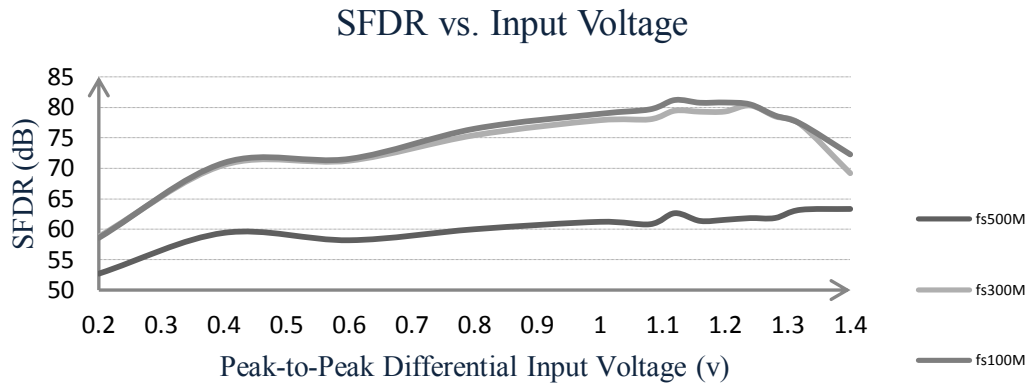


Figure 5-6: SFDR vs. Peak-to-Peak Differential Input Voltage Plot for Ideal ADC with Transistor Level OpAmp

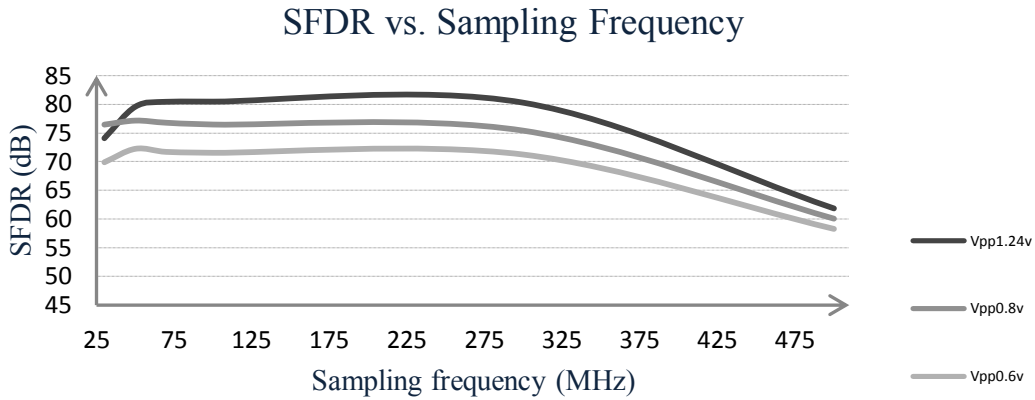


Figure 5-7: SFDR vs. Sampling Frequency Plot for Ideal ADC with Transistor Level OpAmp

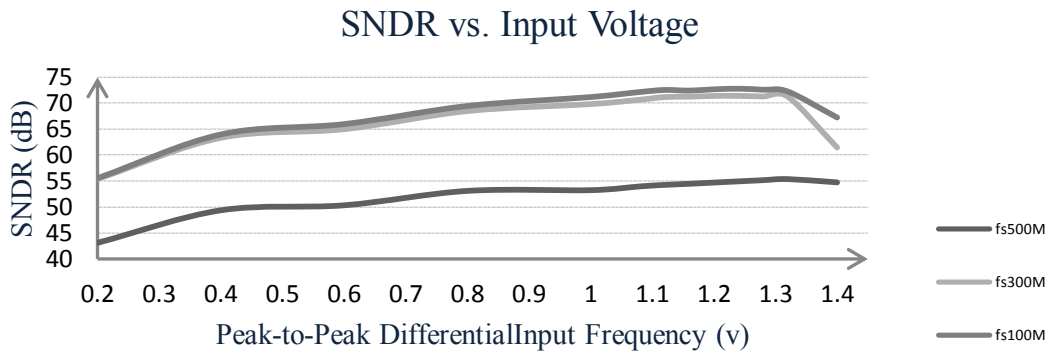


Figure 5-8: SNDR vs. Peak-to-Peak Differential Input Voltage Plot for Ideal ADC with Transistor Level OpAmp

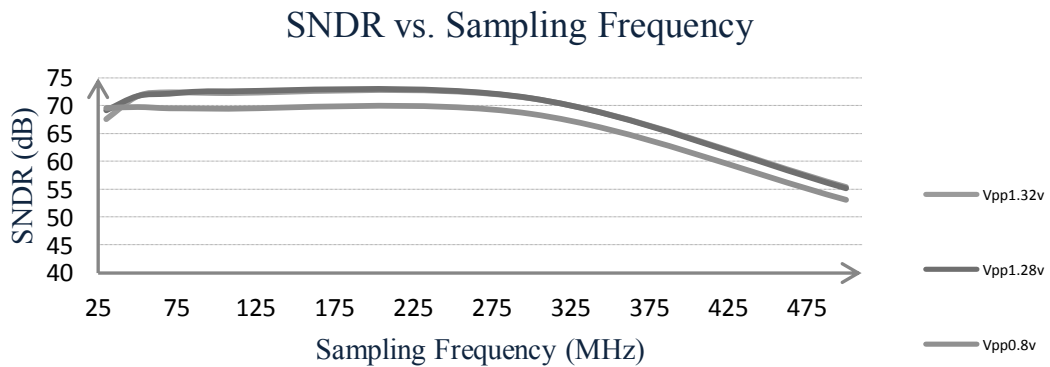


Figure 5-9: SNDR vs. Sampling Frequency Plot for Ideal ADC with Transistor Level OpAmp

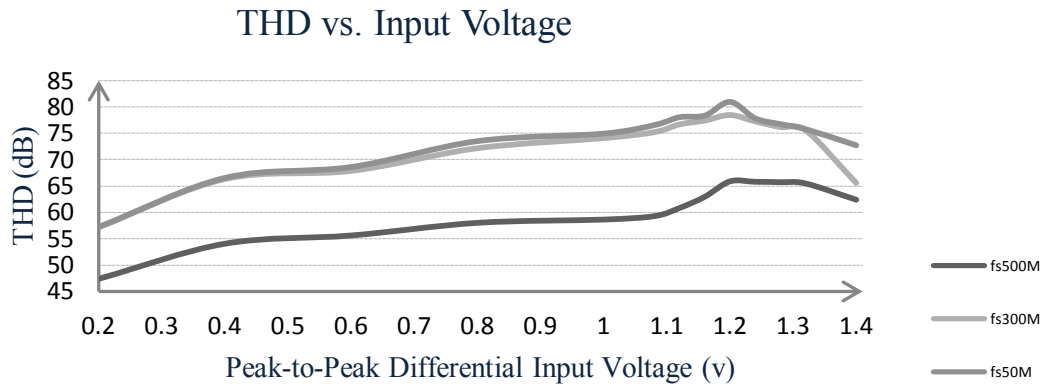


Figure 5-10: THD vs. Peak-to-Peak Differential Input Voltage Plot for Ideal ADC with Transistor Level OpAmp

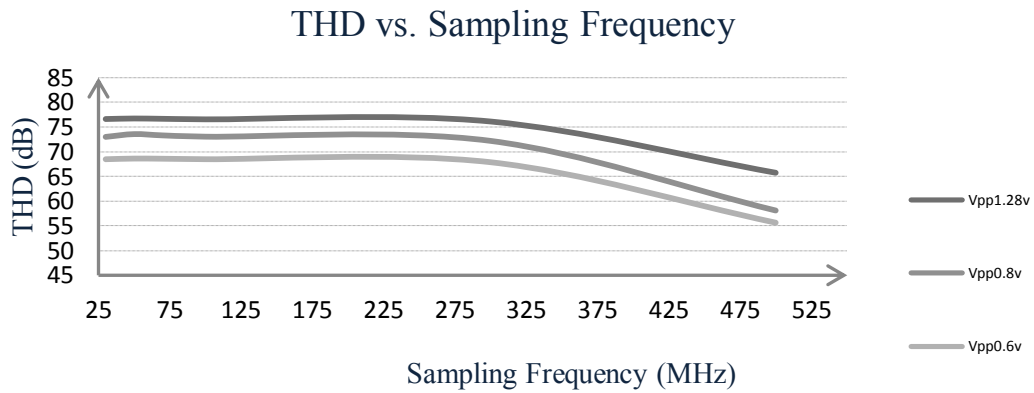


Figure 5-11: THD vs. Sampling Frequency Plot for Ideal ADC with Transistor Level OpAmp

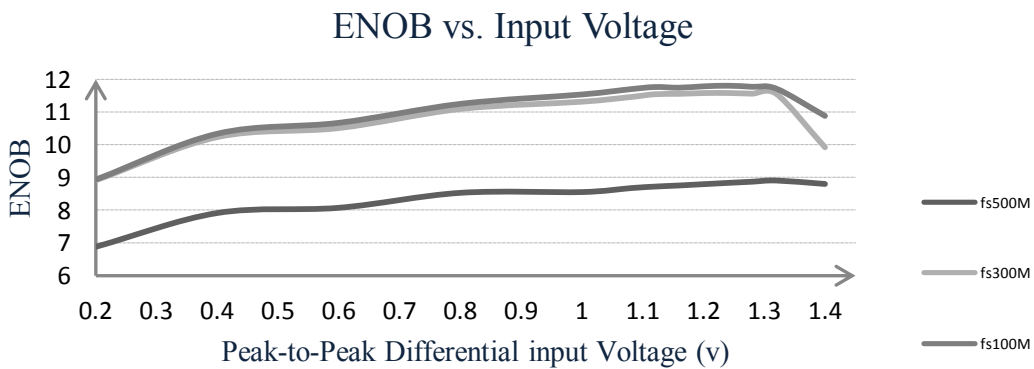


Figure 5-12: ENOB vs. Peak-to-Peak Differential Input Voltage Plot for Ideal ADC with Transistor Level OpAmp

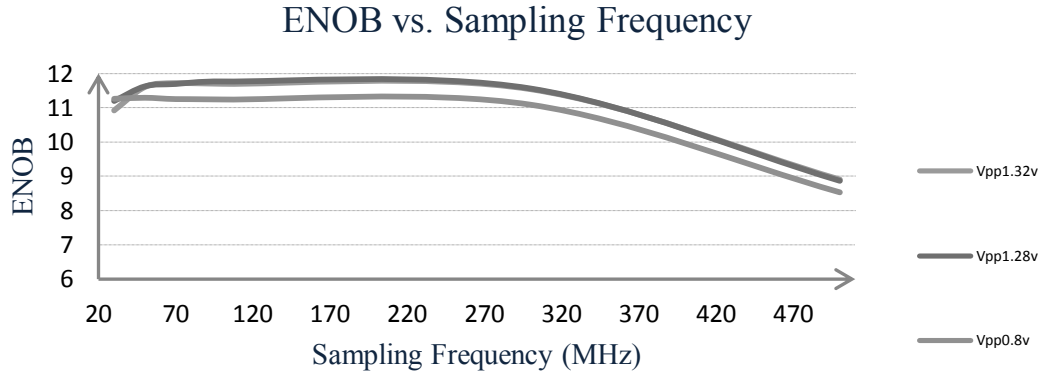


Figure 5-13: ENOB vs. Sampling Frequency Plot for Ideal ADC with Transistor Level OpAmp

The results show that the performance of the ADC improves by increasing the input voltage amplitude up to the point where the distortion in the circuit becomes an issue and degrades the performance. The increase in the input voltage leads to stronger signal power compared to noise and distortion power. This increase, also, leads to more distortion in the circuit, but the increase in signal power is more influential compared to the increase in distortion power, up to the changing point (here 1.32v). Drain current of a CMOS transistor which is biased in saturation can be calculated using this formula:

$$\text{Equation 5-1: } i_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (v_{GS} - v_{TH})^2 (1 + \lambda v_{DS})$$

Where $v_{GS} = V_{Bgs} + v_{gs}$ (dividing gate-source into its biasing voltage and small signal voltage). Therefore, $i_D = I_D + i_d$. Applying Taylor expansion to the current equation around its DC operating point (Q) gives:

$$\text{Equation 5-2: } i_D = I_D + \left. \frac{\partial i_D}{\partial v_{GS}} \right|_Q (v_{gs}) + \frac{1}{2} \left. \frac{\partial^2 i_D}{\partial v_{GS}^2} \right|_Q (v_{gs})^2 + \dots$$

For very small-signal voltages, where the biasing point of the transistor is not perturbed, the drain current can be approximated as:

$$\text{Equation 5-3: } i_D = I_D + \left. \frac{\partial i_D}{\partial v_{GS}} \right|_Q (v_{gs})$$

Where the partial derivative $\left. \frac{\partial i_D}{\partial v_{GS}} \right|_Q$ is called g_m . With this linear behaviour of the

transistors, one can expect that by increasing the input signal the signal to noise and distortion ratio will increase which is the case for small signal operation. However, if the

signal changes are not small then the linear behaviour of transistors is out of the picture and higher order terms in Equation 5-2 cannot be neglected anymore. These terms are the source of the harmonic distortion components and the cause of, for example, SNDR drop for larger input signals.

The results also show that the increase in the sampling frequency tends to degrade the performance as there is less time dedicated to amplification process which means that there is less time for OpAmp to settle within its final value.

In figures above from Figure 5-4 to Figure 5-13 , it can be seen that the ENOB stays around 11.5 bit for sampling frequencies up to 320 MHz. The ENOB drops by less than 1 bit when the OpAmp is placed in the ideal pipelined ADC's circuit. The SNR value is evaluated to be more than 73 dB for f_s up to 320 MHz.

Therefore, the OpAmp can provide an inter-stage gain of 4, in 2.5bps architecture of 5 stages of the 12-bit pipelined ADC without introducing noise, distortion and gain error beyond the tolerance of a high resolution pipelined ADC. This means that the stringent requirements for the OpAmp, to be employed in a 12-bit pipelined ADC, are satisfied in the designed OpAmp.

For further studies all the blocks in the pipelined ADC are designed in the schematic. The designed circuits are shown in Chapter2. The simulation result for the schematic pipelined ADC can be found in Appendix A. This part of the design is beyond the scope of this thesis and obviously there is room for more improvements to be made to the designed blocks. Nevertheless, the experience and knowledge that can be achieved by designing a complete pipelined ADC in schematic is worth the time and the effort.

Future Work

In this work, an OpAmp with very high gain-bandwidth, high linearity and Signal-to-Noise ratio has been designed. The performance of the OpAmp is verified using Cadence simulation and Matlab and they satisfy the requirements on the amplifier of a 2.5bps MDAC in a 12-bit pipelined ADC. The amplifier is placed in a pipelined ADC which is also designed in transistor level. The main focus in this work was the OpAmp design to meet the high requirements needed for 2.5 bps MDAC and provide an inter-stage gain of 4 in the ADC. The OpAmp should provide a high closed-loop bandwidth to accommodate a high speed ADC with very low gain error to match the high resolution definition. However, there are a lot of aspects in a pipelined ADC which could not be covered within the scope of this thesis work.

The most important aspect of a well-designed ADC is its low power consumption. There are several techniques that can be used to reduce the power consumption. For Example, later stages in pipelined ADC can have lower resolution which means lower inter-stage gain and therefore lower power consumption. Not only the lower inter-stage gain will come to assist in this matter but the number of flash Comparators will be reduced which also means lower power consumption as well. As explained in Chapter2, in order to decide on the first stage's resolution the linearity of the ADC should be considered as an important result of this decision as well as the noise reduction by the inter-stage gain; but the later stages have more relaxed requirements regarding their speed and resolution which means their errors due to noise, gain and offset is less effective in the total input referred noise. Due to the mentioned fact, scaling is another option that helps with the reduction of power consumption, specially scaling the OpAmp which is the most power hungry part of the ADC.

Error sources in an ADC can be dealt with after data conversion in digital domain or in analogue domain via calibration techniques [20]. Choosing either one of them (or both of them) there are lots of approaches developed during years of research that can be used. Of course, there is always room for further development to find new approaches or improve existing ones. Calibration techniques which work in analogue domain tend to increase the circuit complexity and power consumption, making the digital calibration techniques more interesting. In digital domain, calibration techniques fall into two categories [20]: Foreground Calibration and Background Calibration techniques. Foreground calibration can be done when the ADC is in calibration mode which means the ADC cannot work normally during calibration time. During calibration process a test signal is applied to the ADC and the result is compared against ideal response. The error is subtracted from the ADC's digital outputs. On the other hand, background calibration circuitry calibrates the ADC as it does its normal operation and they are able to monitor and control the errors due to circuit and environmental change due to time and temperature and etc. Choosing the suitable calibration technique for the application and modifying and improving the technique could be an interesting and important field of research.

Time interleaving is another very interesting area of research. The idea is to employ K ADCs in parallel with a sampling frequency of $\frac{1}{K}$ total sampling frequency of the converter. The output is obtained by multiplexing between the individual ADCs' output bits at a sample rate of K times of their sampling frequency. This way, by incorporating K

individual ADCs with sampling frequency of f_s , total sampling frequency would be $K f_s$. Although, very high sampling rates (tens of GS/s) can be achieved by this approach, the resolution is mostly limited to 8 bits. The problem is that the matching requirements in K channels are very high and existing methods such as matching physical channel layouts, using common ADC reference voltages, pre-screening devices, and active analogue trimming can help with resolutions around 8 bits [21]. But for higher resolutions, stringent matching requirements cannot be satisfied using these methods. Overcoming Matching problem to achieve higher resolution ADCs is a new and wide area of research to explore.

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Appendix A

Simulation Result for the Pipelined ADC in Transistor Level

All Blocks in the 5 stages of the pipelined ADC plus the back-end flash ADC are designed in transistor level. SNR, SFDR, SNDR, THD and ENOB are calculated and the simulation results can be seen and compared with each other in respect to sampling frequency and input amplitude changes in coming figures.

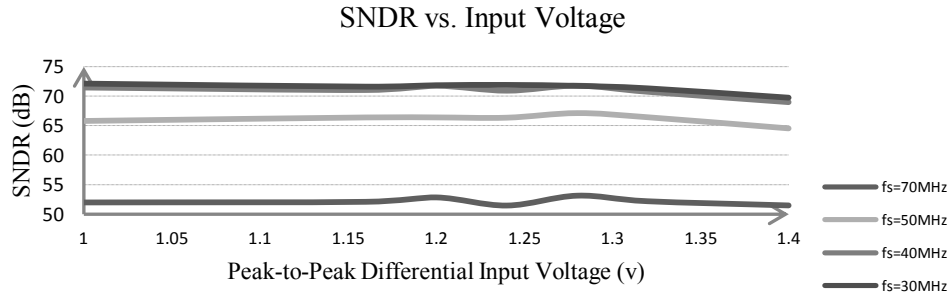


Figure 0-1: SNDR vs. Peak-to-Peak Differential Input Voltage Plot for Transistor Level Pipelined ADC

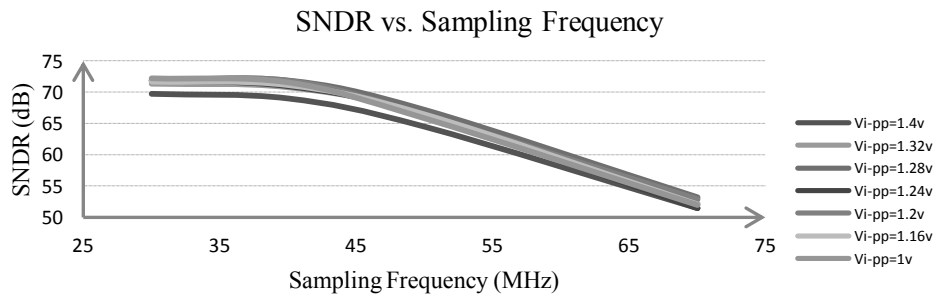


Figure A-2: SNDR vs. Sampling Frequency Plot for Transistor Level Pipelined ADC

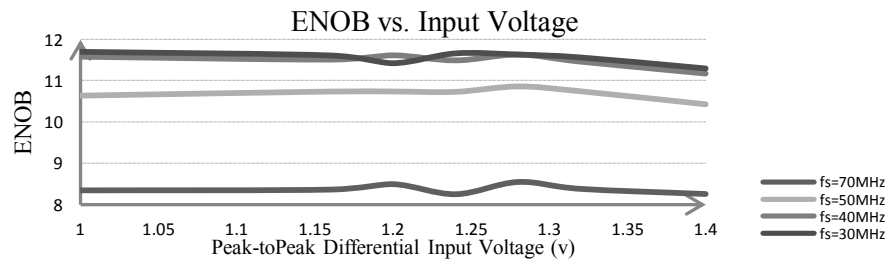


Figure 0-3: ENOB vs. Peak-to-Peak Differential Input Voltage Plot for Transistor Level Pipelined ADC

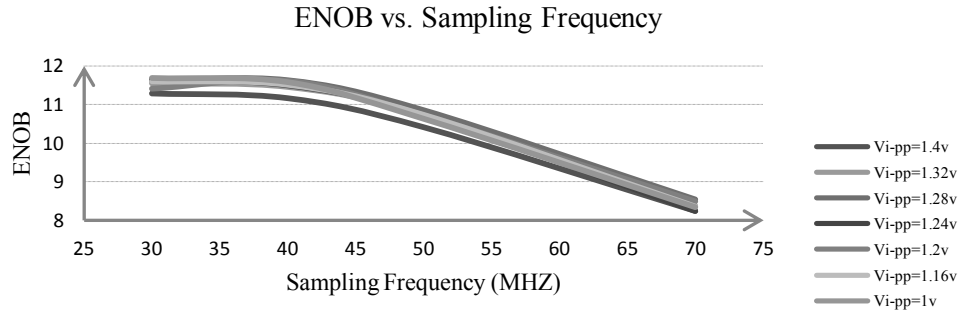


Figure 0-4: ENOB vs. Sampling Frequency Plot for Transistor Level Pipelined ADC

The results show that the SNDR of the Schematic pipelined ADC stays above 70dB maintaining an ENOB more than 11 up to 45MHz sampling frequency. Sampling frequency and input voltage changes affect the performance of the ADC in a similar way as previous two ADC models studied in Chapter 5. The only difference for completely schematic model is the highest sampling frequency that can be applied to the circuit before the performance falls below system specification. The clocking scheme which is described in Chapter 2 leaves shorter time for amplification phase during the sampling clock period. Therefore a longer clock period is needed.

In fully schematic pipelined ADC, other parts of the circuit apart from the OpAmp introduce error to the system and degrade the performance. Some examples to be named here are comparator's offset, nonlinearity; clock feed-through and charge injection of sampling switches in switched capacitor sampling network, etc. More details can be found in Chapter 2.

Appendix B

VerilogA Codes

VerilogA Code for 12-bit Digital Writer

```
`include "constants.vams"

`include "disciplines.vams"

// log_time = if time should be dumped in the text file; 1=>yes, 0=>no

// Samples 12 input voltages every rising edge of the clk and writes the results to the file
//'filename'
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////

module Writer_Digital(clk,Vin);

input clk;

electrical clk;

input [11:0] Vin;

electrical [11:0] Vin;

////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////Parameters

parameter real vtrans_clk = 0.55 ;

parameter integer log_time=1;

parameter string fileName = "/Address/File Name.txt";

////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////Variables

integer outputFileId;

////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////

analog begin

@(initial_step)

begin

outputFileId = $fopen(fileName);
```

```

end

@ (cross(V(clk) - vtrans_clk, 1)) begin

    if (log_time == 1 ) begin

        $fwrite(outputFileId, "%f %f %f %f %f %f %f %f %f %f %f %f %f\n", $abstime*1e9,
        V(Vin[11]),V(Vin[10]),V(Vin[9]),V(Vin[8]),V(Vin[7]),V(Vin[6]),V(Vin[5]),V(Vin[4]),V(
        Vin[3]),V(Vin[2]),V(Vin[1]),V(Vin[0]));

    end

    else begin

        $fwrite(outputFileId, "%ft%ft%ft%ft%ft%ft%ft%ft%ft%ft%ft%ft%ft\n", V(Vin[11]),
        V(Vin[10]),V(Vin[9]),V(Vin[8]),V(Vin[7]),V(Vin[6]),V(Vin[5]),V(Vin[4]),V(Vin[3]),V(
        Vin[2]),V(Vin[1]),V(Vin[0]));

    End      end

@ ( final_step ) begin

    $fclose(outputFileId);

End      endmodule

```

VerilogA Code for Differential Analogue Writer

```

`include "constants.vams"

`include "disciplines.vams"

// log_time = if time should be dumped in the text file; 1=>yes, 0=>no

// Samples an Analogue input every rising edge of the clk and writes the results to the file
//'filename'

////////////////////////////////////

module Writer_Analogue(clk, Vip, Vin);

    input clk;

    electrical clk;

    input Vip, Vin;

    electrical Vip, Vin;

```

```

////////////////////////////////////Parameters

parameter real vtrans_clk = 0.55 ;

parameter integer log_time=1;

parameter string fileName = "/Address/File Name.txt";

////////////////////////////////////Variables

integer outputFileId;

real Sample;

integer i;

////////////////////////////////////

    analog begin

    @(initial_step)

    begin

        outputFileId = $fopen(fileName);

        i=0;

    end

    @ (cross(V(clk) - vtrans_clk, 1))

    begin

        Sample= V(Vip)-V(Vin);

        i=i+1;

        if (log_time == 1 ) begin

            $fwrite(outputFileId, "%d\t%-.10g\t%f\n",i,$abstime*1e9,Sample);

        end

    else begin

        $fwrite(outputFileId, "%d\t%f\n" ,i,Sample);

```

```

end      end

@ ( final_step ) begin

$fclose(outputFileId);

End      end      endmodule

```

VerilogA Code for Differential 16-bit Scalable DAC

```

`include "constants.vams"

`include "disciplines.vams"

module DAC_16bit_Scalable(DACout, DACoutp,DACoutn, DACin0p, DACin1p,
DACin2p, DACin3p, DACin4p, DACin5p, DACin6p, DACin7p, DACin8p, DACin9p,
DACin10p, DACin11p, DACin12p, DACin13p, DACin14p, DACin15p, DACin0n,
DACin1n, DACin2n, DACin3n, DACin4n, DACin5n, DACin6n, DACin7n, DACin8n,
DACin9n, DACin10n, DACin11n, DACin12n, DACin13n, DACin14n, DACin15n,
DacCLK);

output DACout, DACoutp,DACoutn,

electrical DACout, DACoutp,DACoutn,

input DACin0p,DACin0n, DACin1p,DACin1n, DACin2p,DACin2n, DACin3p,DACin3n,
DACin4p,DACin4n, DACin5p,DACin5n, DACin6p,DACin6n, DACin7p,DACin7n,
DACin8p,DACin8n, DACin9p,DACin9n, DACin10p,DACin10n, DACin11p,DACin11n,
DACin12p,DACin12n, DACin13p,DACin13n, DACin14p,DACin14n,
DACin15p,DACin15n;

electrical DACin0p,DACin0n, DACin1p,DACin1n, DACin2p,DACin2n,
DACin3p,DACin3n, DACin4p,DACin4n, DACin5p,DACin5n, DACin6p,DACin6n,
DACin7p,DACin7n, DACin8p,DACin8n, DACin9p,DACin9n, DACin10p,DACin10n,
DACin11p,DACin11n, DACin12p,DACin12n, DACin13p,DACin13n,
DACin14p,DACin14n, DACin15p,DACin15n;

input DacCLK;  electrical DacCLK;

////////////////////////Parameters

parameter real tfall = 10p from [0:inf) ;

parameter real trise = 10p from [0:inf) ;

parameter real tdel = 1p  from [0:inf) ;

parameter real Vrefp = 900m ;

```



```

parameter real Vrefn = 200m ;

parameter real vtrans = 0.55 ;

parameter real vtrans_clk = 0.55 ;

parameter real vlogic_low = 0.0 ;

parameter real vlogic_high = 1.1 ;

parameter real DAC_Res=16;

////////////////////Variables

integer DigitalInMapp[15:0],integer DigitalInMapn[15:0],integer Step,integer i;

real AccumulatorDACp, AccumulatorDACn,real AnalogueDAC, AnalogueDACp,
AnalogueDACn,real weight;

////////////////////////////////////

analog begin

@(initial_step) begin

weight = (Vrefp-Vrefn)/(pow(2,DAC_Res)-1);

$strobe("ssssssssssssssssssssssssssssssssssssssssssssssssssssss_weightDAC_Diff %e\n",weight);

i=0;      Step=0;      AccumulatorDACp=0;      AccumulatorDACn=0;

AnalogueDAC =0;      AnalogueDACp=0;      AnalogueDACn=0;

end

@(cross( V(DacCLK) - vtrans_clk , +1 ))begin

if (V(DACin0p)>vtrans)   DigitalInMapp[0]=1; else   DigitalInMapp[0]=0;

if (V(DACin1p)>vtrans)   DigitalInMapp[1]=1; else   DigitalInMapp[1]=0;

if (V(DACin2p)>vtrans)   DigitalInMapp[2]=1; else   DigitalInMapp[2]=0;

if (V(DACin3p)>vtrans)   DigitalInMapp[3]=1; else   DigitalInMapp[3]=0;

if (V(DACin4p)>vtrans)   DigitalInMapp[4]=1; else   DigitalInMapp[4]=0;

if (V(DACin5p)>vtrans)   DigitalInMapp[5]=1; else   DigitalInMapp[5]=0;

```

```

if (V(DACin6p)>vtrans)  DigitalInMapp[6]=1; else    DigitalInMapp[6]=0;
if (V(DACin7p)>vtrans)  DigitalInMapp[7]=1; else    DigitalInMapp[7]=0;
if (V(DACin8p)>vtrans)  DigitalInMapp[8]=1; else    DigitalInMapp[8]=0;
if (V(DACin9p)>vtrans)  DigitalInMapp[9]=1; else    DigitalInMapp[9]=0;
if (V(DACin10p)>vtrans) DigitalInMapp[10]=1; else   DigitalInMapp[10]=0;
if (V(DACin11p)>vtrans) DigitalInMapp[11]=1; else   DigitalInMapp[11]=0;
if (V(DACin12p)>vtrans) DigitalInMapp[12]=1; else   DigitalInMapp[12]=0;
if (V(DACin13p)>vtrans) DigitalInMapp[13]=1; else   DigitalInMapp[13]=0;
if (V(DACin14p)>vtrans) DigitalInMapp[14]=1; else   DigitalInMapp[14]=0;
if (V(DACin15p)>vtrans) DigitalInMapp[15]=1; else   DigitalInMapp[15]=0;

////////////////////////////////////
///

if (V(DACin0n)>vtrans)  DigitalInMapn[0]=1; else    DigitalInMapn[0]=0;
if (V(DACin1n)>vtrans)  DigitalInMapn[1]=1; else    DigitalInMapn[1]=0;
if (V(DACin2n)>vtrans)  DigitalInMapn[2]=1; else    DigitalInMapn[2]=0;
if (V(DACin3n)>vtrans)  DigitalInMapn[3]=1; else    DigitalInMapn[3]=0;
if (V(DACin4n)>vtrans)  DigitalInMapn[4]=1; else    DigitalInMapn[4]=0;
if (V(DACin5n)>vtrans)  DigitalInMapn[5]=1; else    DigitalInMapn[5]=0;
if (V(DACin6n)>vtrans)  DigitalInMapn[6]=1; else    DigitalInMapn[6]=0;
if (V(DACin7n)>vtrans)  DigitalInMapn[7]=1; else    DigitalInMapn[7]=0;
if (V(DACin8n)>vtrans)  DigitalInMapn[8]=1; else    DigitalInMapn[8]=0;
if (V(DACin9n)>vtrans)  DigitalInMapn[9]=1; else    DigitalInMapn[9]=0;
if (V(DACin10n)>vtrans) DigitalInMapn[10]=1; else   DigitalInMapn[10]=0;
if (V(DACin11n)>vtrans) DigitalInMapn[11]=1; else   DigitalInMapn[11]=0;
if (V(DACin12n)>vtrans) DigitalInMapn[12]=1; else   DigitalInMapn[12]=0;

```

```

if (V(DACin13n)>vtrans) DigitalInMapn[13]=1; else    DigitalInMapn[13]=0;
if (V(DACin14n)>vtrans) DigitalInMapn[14]=1; else    DigitalInMapn[14]=0;
if (V(DACin15n)>vtrans) DigitalInMapn[15]=1; else    DigitalInMapn[15]=0;

////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
////

i=0;    Step=0;    AccumulatorDACp=0;    AccumulatorDACn=0;
AnalogueDAC =0;    AnalogueDACp=0;    AnalogueDACn=0;
for (i=0; i<DAC_Res; i=i+1)begin
    Step = pow(2,i);
    if (DigitalInMapp[i]==1)begin
        AccumulatorDACp = AccumulatorDACp + (DigitalInMapp[i] * Step);
    end
    if (DigitalInMapn[i]==1)begin
        AccumulatorDACn = AccumulatorDACn + (DigitalInMapn[i] * Step);
    end
end

AnalogueDACp = (AccumulatorDACp * weight )+ Vrefn;
AnalogueDACn = (AccumulatorDACn * weight )+ Vrefn;
AnalogueDAC = (AnalogueDACp - AnalogueDACn)+ vtrans;
end

V(DACoutp) <+ transition ( AnalogueDACp , tdel, trise, tfall);
V(DACoutn) <+ transition ( AnalogueDACn , tdel, trise, tfall);
V(DACout)  <+ transition ( AnalogueDAC  , tdel, trise, tfall);

end    endmodule

```


Matlab Codes

Matlab Code for Reading Text File from Cadence for OpAmp

```
%% file reading from cadence& extracting sampled data from file for OpAmp
y= textread('Address\File Name.txt','%f');
A= size(y);
i=3:3:A(1);%row 3 of the input file
Sample=y(i);
%% Initial conditions
fin    = Input Frequency;
fs     = Smpling Frequency;
Vmax   = Peak voltage of single input signal;
Vpp    = Output peal-to-peak Differential voltage;
OSR    = fs/(2*fin);
BW     = fin+0.01e3;
Ms     = 4096; %%2^12 sample = FFT's Size,
Vtrans = 0.55; %% Common-Mode Voltage
DAC_Res= 12;
%% Excluding initial samples in case of not settled signal
B=size(Sample);
j=(B(1)-4095):1:B(1); %% 4096 last samples
Sample_LPfiltered = Sample (j);
fprintf('-----\n');
fprintf('fin = %1.3f (MHz),\t', fin/1e6);
fprintf('fs = %1.3f (GHz),\t', fs/1e9);
fprintf('Vpp= %1.3f (mv)\n', Vpp*1e3);
fprintf('-----\n');
%% Function call to calculate performance metrics
SNDR(Sample_LPfiltered,Ms,fin,fs,OSR,Vpp);
SNDRsine(Sample_LPfiltered,fin,fs,Ms,BW);
```

Matlab Code for Reading Text File from Cadence for ADC

```
%file reading from cadence; Reading x-bit(ADC's digital outputs) digital value+ time for
%each
% Word and calling DAC function to provide the analogue value for each word
% DAC_Res=12 ;==> DAC_Res+1 covers first column for time and other columns for
%digital voltages.
% Remember to subtract the one if log_time in cadence is zero.
% A=size (yp);==>K = A(1)/(DAC_Res+1)
% X1 = reshape (yp, DAC_Res+1, K); reshape reads vertically and goes DAC_res times
%down then comes back to the next column==> size=(DAC_Res+1,K)
% X2 gives X1 but in correct size (K, DAC_Res+1) ==>next line ignores first column
%(time).
%% Dip and Din are digital data lines from ADC in the order (D11 D10 ...D0)
%==> For DAC function to operate correctly we have to rearrange columns to
% have this order (D0 D1 ... D11) in 2 other matrixes Dip1 and Din1 ==> the order can be
% changed by changing the write order in cadence
%% From Pipelined_ADC 's outputs
```

```

yp = textread('Address/File Name.txt','%f');
yn = textread('Address/File Name.txt','%f');
%% Initial conditions
fin    = Input Frequency;
fs     = Smpling Frequency;
Vmax   = Peak voltage of single input signal;
Vpp    = Output peal-to-peak Differential voltage;
OSR    = fs/(2*fin);
BW     = fin+0.01e3;
Ms     = 4096; %%2^12 sample = FFT's Size,
Vtrans = 0.55; %% Common-Mode Voltage
DAC_Res= 12;
S1=size(yp);
k1=S1(1)/(DAC_Res+1);
X1 = reshape(yp,(DAC_Res+1),k1);
X2 = X1';
Dip    = X2(2:k1,2:(DAC_Res+1));
Dip1 = [Dip(:,12),Dip(:,11),Dip(:,10),Dip(:,9),Dip(:,8),Dip(:,7),Dip(:,6),Dip(:,5),Dip(:,4),
Dip(:,3),Dip(:,2),Dip(:,1)];
S2=size(yn);
k2=S2(1)/(DAC_Res+1);
Y1 = reshape(yn,(DAC_Res+1),k2);
Y2 = Y1';
Din    = Y2(2:k2,2:(DAC_Res+1));
Din1 = [Din(:,12),Din(:,11),Din(:,10),Din(:,9),Din(:,8),Din(:,7),Din(:,6),Din(:,5),
Din(:,4),Din(:,3), Din(:,2),Din(:,1)];
%% Calling DAC function
AnalogueDAC_Diff=DAC(Dip1,Din1,Vtrans,Vmax,DAC_Res);
plot(AnalogueDAC_Diff,'--b');
%% excluding initial samples in case of not settled signal, considering only Ms last
samples
Sample = AnalogueDAC_Diff;
B=size(Sample);
j=(B(1)-(Ms-1)):1:B(1);
Sample_LPfiltered=Sample(j);
figure(4);stem(Sample_LPfiltered)
xlabel('N','fontsize',14,'fontweight','b')
ylabel('Sample_LPfiltered,amp(mv)','fontsize',14,'fontweight','b')
fprintf('-----\n');
fprintf('fin = %1.3f (MHz),\t', fin/1e6);
fprintf('fs = %1.3f (MHz),\t', fs/1e6);
fprintf('Vpp = %1.3f (mv)\n', Vpp*1e3);
fprintf('-----\n');
%% Function call to calculate performance metrics
SNDR(Sample_LPfiltered,Ms,fin,fs,OSR,Vpp);
SNDRsine(Sample_LPfiltered,fin,fs,Ms,BW);

```

Matlab DAC Code for Reconstructing Digital Outputs of the ADC

```

function AnalogueDAC_Diff = DAC(Dip,Din,Vtrans,Vmax,DAC_Res)
Vrefp = Vtrans+Vmax;
Vrefn = Vtrans-Vmax;
weight = (Vrefp-Vrefn)/((2^DAC_Res)-1);
Sp = size(Dip);
Sn = size(Din);
Stepp=0;
Stepn=0;
AccumulatorDACp =zeros(1,Sp(1));
AccumulatorDACn =zeros(1,Sp(1));
AnalogueDAC_p =zeros(1,Sp(1));
AnalogueDAC_n =zeros(1,Sp(1));
Dp =zeros(Sp(1),DAC_Res);
Dn =zeros(Sp(1),DAC_Res);
%%%%%%%%%%%%%%
for i=1:1:Sp(1);
    for j=0:1:DAC_Res-1
        if (Dip(i,j+1)>Vtrans)
            Dp(i,j+1)=1;
        else
            Dp(i,j+1)=0;
        end
        if (Dp(i,j+1)==1)
            Stepp = 2^j;
            AccumulatorDACp(i) = AccumulatorDACp(i) + (Dp(i,j+1) * Stepp);
        end
    end
    AnalogueDAC_p(i) = (AccumulatorDACp(i) * weight );
end
%%%%%%%%%%%%%%
for m=1:1:Sn(1);
    for n=0:1:DAC_Res-1
        if (Din(m,n+1)>Vtrans)
            Dn(m,n+1)=1;
        else
            Dn(m,n+1)=0;
        end
        if (Dn(m,n+1)==1)
            Stepn = 2^n;
            AccumulatorDACn(m) = AccumulatorDACn(m) + (Dn(m,n+1) * Stepn);
        end
    end
    AnalogueDAC_n(m) = (AccumulatorDACn(m) * weight );
end
%%%%%%%%%%%%%%
Vcm=ones(Sp(1),1)*Vtrans;
AnalogueDAC_p=reshape(AnalogueDAC_p,Sp(1),1);
AnalogueDAC_n=reshape(AnalogueDAC_n,Sn(1),1);
AnalogueDAC_Diff = AnalogueDAC_p - AnalogueDAC_n;

```

```

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
t = 1:1:Sp(1);
subplot(2,1,2);
figure(1);plot(t,AnalogueDAC_p,'-r')
xlabel('N','fontsize',14,'fontweight','b')
ylabel('DAC_amp(mv)','fontsize',14,'fontweight','b')
hold on
plot(t,AnalogueDAC_n,'-.m')
plot(t,AnalogueDAC_Diff,'--b')
hold off
end

```

Matlab Code for Calculation of Performance Metrics of ADC and OpAmp¹

```

% N number of points in FFT
% fin input frequency
% fs sampling frequency
% osr = fs/(2*BW); BW is fin in most cases
% Va signal amplitude for FFT normalisation
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
function [SNR,SNDR] = SNDR(Vin,N,fin,fs,osr,Va)
%FFT plot of input signal
Vinfft=2*abs(fft(Vin))/(N*Va);
plot((1:1:length(Vinfft)),20*log10(Vinfft))
Vinfft=Vinfft(1:N/2); % Dumping half of the Bandwidth to consider just Nyquist Band
Sp = 20 *log10(Vinfft(round(N*fin/fs+1))); %Signal Power
Vinfft(round(N*fin/fs+1)) = 0; %%Neglecting input component in FFT to calculate noise
and distortion power
Vinfft(1) = 0; %%Neglecting the DC
NDp = 10 *log10(sum(Vinfft.^2));%% Noise and Distortion Power
THD= Sp - (10
*log10((Vinfft(round(N*3*fin/fs+1)).^2)+(Vinfft(round(N*5*fin/fs+1)).^2)
+(Vinfft(round(N*7*fi/fs+1)).^2)+(Vinfft(round(N*9*fin/fs+1)).^2)+(Vinfft(round(N*11*
fi/fs+1)).^2)+(vinfft(round(N*13*fin/fs+1)).^2)+(Vinfft(round(N*15*fin/fs+1)).^2)+(Vinfft
t(round(N*17*fin/fs+1)).^2)));
SNDR = Sp - NDp;
ENOB = (SNDR-1.76)/6.02;
%%SFDR
Max-Spur= abs(Vinfft(2));
for i=2:1:N/2
    if (Max-Spur< abs(Vinfft(i)))
        Max-Spur = abs(Vinfft(i));
    end
end
Max-Spur = 10 *log10(Max-Spur^2);
SFDR = Sp – Max-Spur;
Vinfft(round(N*3*fin/fs+1)) = 0;

```

¹ This code is written based on a code by Ameya Bhide, current Ph.D. student in Linköping university


```

Vinfft(round(N*5*fin/fs+1)) = 0;
Vinfft(round(N*7*fin/fs+1)) = 0;
Vinfft(round(N*9*fin/fs+1)) = 0;
Np = 10 *log10(sum(Vinfft.^2));%%Noise Power
SNR= Sp - Np;
fprintf('-----\n');
fprintf('SignalPower   = %1.3f dB\n', Sp);
fprintf('NoisePower    = %1.3f dB\n', Np);
fprintf('N&D Power     = %1.3f dB\n', NDp);
fprintf('-----\n');
fprintf('THD           = %1.3f dB\n', THD);
fprintf('SNR           = %1.3f dB\n', SNR);
fprintf('SFDR          = %1.3f dB\n', SFDR);
fprintf('SNDR          = %1.3f dB\n', SNDR);
fprintf('-----\n');
fprintf('ENOB          = %1.3f \n', ENOB);
fprintf('-----\n');

```

Matlab Code for Calculation of Performance Metrics of ADC and OpAmp²

```

%% this function builds an ideal sine wave using input and sampling frequency and
%% subtracts the actual signal from ideal one to obtain an error signal which is assumed to
%% be due to noise and distortion.
%% BW is chosen to be slightly larger than input signal to cover the fundamental signal.
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
function SINAD = SNDRsine(InData,fin,fs,N,BW)
output_raw = InData';
output_size = size(output_raw);
%% this depends on the input data, whether it is vertical matrix or horizontal, next 2
lines are not necessary here as their action has been taking care of in file read code
% out = output_raw(output_size(1)-(N-1):output_size(1)); % Use only the N last samples
out = output_raw(output_size(2)-(N-1):output_size(2)); % Use only the N last samples
Fc=fs/(2*bw);
%%Constructing ideal sine wave according to fin and fs
w = 2*pi*fin/fs;
t = 1:1:length(out);
D_null = [cos(w*t);sin(w*t);ones(size(t))];
x_null = inv(D_null*D_null)*D_null*out';
out_fitted = (D_null*x_null)';
e = (out-out_fitted); % Error between input sine wave and ideal sine wave
rms_signal = sqrt((x_null(1)^2+x_null(2)^2)/2);
rms_noise = sqrt(sum(e.*e)/N);
sinad = 20*log10(rms_signal/rms_noise);
enob=(sinad-1.76)/6.02;
fprintf('Performance Estimation with Sine Extraction\n\n');
fprintf('SNDR(sine) = %1.3f dB\n', sinad);
fprintf('ENOB      = %1.3f \n', enob);

```

² This code is a modified version of a code by Timmy Sundstrom, former Ph.D. student in Linköping university, to compare the result with frequency domain approach